

Closed lambda terms (OEIS A135501 and A220894)

All logarithmic orders, a controlled threshold inverse,
and the random binding geometry of a closed term

Two research reports of one external research session
(bundle Reports 110 and 109), merged into one ProveIt report

Parts I and II: 2 October 2026 merged: 5 October 2026

Abstract

Let $A_s(n)$ count the closed lambda terms of size n up to alpha-equivalence, when abstractions and applications have size one and variable occurrences size $s \in \{0, 1\}$; A_1 is OEIS A135501 and A_0 is OEIS A220894. Put $r = s + 1$, $N = n + 1$, $t = W(4eNe^{-r/2})$ and $M_r(N) = \frac{N}{r}(t - 2 + t^{-1}) + \frac{N}{2}$. This report prints two manuscripts as Parts I and II. Part I proves $M_r(N) - O(\log N) \leq \log A_s(n) \leq M_r(N) + O(N^{1-1/(3r)}(\log N)^{-2/3+1/(3r)})$, hence every fixed inverse-logarithmic order of $\log A_s(n)$ (in particular the coefficient $\frac{1}{2} - \frac{\log 4}{r}$ of $n/\log n$), with polynomial recurrences for all orders and a controlled threshold inverse with an asymmetric bracket. Part II proves the weaker four-scale form, with error $O(n/\log n)$, by a second route, and a joint large-deviation principle at speed n for the scaled number of abstractions and the maximum unary height of a uniform random closed term, with rate $(t - 1 - \log x)/r$ on $0 < x \leq t$: both scaled parameters tend to 1 in every L^p , and a longest unary path carries all but $o(n/\log n)$ abstractions in probability. A remark added in the merge shows that Part I's estimates settle, to leading order, the ratio $R(b, u) = A(b, u)/(C_b u^{b+1})$ that Part II leaves open: $\log R(b, u)/u \rightarrow \frac{1}{2}$ at the moving saddle. No multiplicative equivalent, amplitude or transseries is proved, and the error exponent $1 - 1/(3r)$ is an upper bound only. The report is unrefereed, and nothing in it is formalized.

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Guide to this report

The two Parts are two manuscripts of the session bundle of Reports 1–243 (arrival commit 60f54ea06), placed in this directory by commit 9c995cefe (batch 103). Both are dated 2 October 2026; Report 109 was written first (its index time is 08:01:33Z, Report 110’s 08:17:00Z), but Report 110 proves strictly stronger counting and inverse theorems under the same hypotheses, so it is the base and comes first:

Part	Bundle report	Delivered title	Labels
I	110 (base)	All logarithmic orders for closed lambda terms	lam:ao:
II	109	Unrestricted closed lambda terms	lam:bg:

Each Part prints its manuscript’s text, with every result, proof, remark, open question and limitation; statements keep their delivered wording. Sections, statements and equations are numbered continuously, and statements and equations are numbered within their sections. Part I keeps Report 110’s section, statement and equation numbers (Sections 1–10), with two exceptions inside Section 3 and one addition: Report 110’s Lemma 3.1 and Report 109’s Theorem 3.1 (upper half) are the same theorem with the same proof, so the proof is printed once, in Report 109’s fuller wording, as Sections 3.1–3.3; Report 109’s Lemmas 3.2 and 3.3 and its Remark 3.4 keep their numbers there, and its equations (3.6)–(3.11) become (3.2)–(3.7). Section 9.2 is added in the write. In Part II a number is Report 109’s plus 10 in its section part: Report 109’s Section k is Section $k + 10$, and its Theorem $k.j$ and equation $(k.j)$ are Theorem $(k + 10).j$ and $(k + 10.j)$. Section 13 keeps only the statement of Report 109’s Theorem 3.1, as Theorem 13.1: Report 109’s (3.2)–(3.5) are Part I’s (2.1)–(2.3) and the display after them, and its (3.6)–(3.11) are in Part I as just said. Part II ends with a section added in the write, Section 20.

Notes added when the manuscripts were merged are set in small type and tagged [**write**], with their date. They give cross-references between the Parts, mark results proved twice, mark statements that Part I supersedes, and record the one deduction that the write adds, Remark 18.1, with its strengthening, Proposition 18.2, added after the independent check of the write (5 October 2026), which proposed its idea; that check is recorded at the end of Section 20. Superseded statements of Part II are printed as delivered, with a dated note pointing to Part I, as in the merged reports **a202058-ascent-000-growth** and **a336070-weak-ascents** of the same collection. No delivered statement was changed. One symbol was renamed, in text moved between the manuscripts (Section 3); no symbol in a Part’s own text was renamed. Each Part ends with a “Further questions and research” section (Sections 9.2 and 20), which collects, under Vladimir’s standing rule of 4 October 2026, every claim of that Part that is stated without proof, with its source, its sketch and what is missing.

What is proved, and where

Here $A_s(n)$ is as in the abstract, $A(b, u) = A_s(n, u)$ counts closed terms with b applications and u abstractions ($n = u + rb + s$), $L = \log N$, and $R_r(N) = N^{1-1/(3r)}(\log N)^{-2/3+1/(3r)}$.

Statement	Where
$M_r(N) - O(\log N) \leq \log A_s(n) \leq M_r(N) + O(R_r(N))$; every fixed inverse-logarithmic order, with the polynomials P_j Weaker: $\log A_s(n) = \frac{N}{r}(w - 2 + \frac{1}{w}) + O(N/\log N)$, $w = W(4eN)$; the four-scale form; $\frac{\log n}{n} A_s(n)^{r/n} \rightarrow \frac{4}{e}$ Threshold inverse $\nu_s(y)$: asymmetric bracket $N_0 - O(R_r(N_0)/\log N_0) \leq \nu_s(y) + 1 \leq N_0 + O(1)$, parametric N_0 , explicit form with error $O(Y(1+k)^2/v^4)$, every fixed inverse order Weaker inverse, error $O(L/\ell^3)$ in Part II's letters Exact recurrences: the binder-context series T_m and the radical recurrence; the size recurrence; the skeleton sandwich $C_b u^{b+1} \leq A(b, u) \leq C_b \binom{u+2b}{u} u^{b+1}$ and the exact weight formula; reduced shape counts $C_q \binom{u+q-1}{2q}$ Coarse majorant $A(b, u) \leq 2uK^u(4u)^b$, $K = e(2 + \sqrt{3})$, with the depth-forest and radical-product lemmas Height-sensitive bound; $\log A(b, u) \leq b \log(4u) + u/2 + E(b, u) + O(\log(u+1))$; exact single-spine series and the Catalan-convolution lower bound $\log R(b, u)/u \rightarrow \frac{1}{2}$ at the moving saddle; an upper bound for all (b, u) and the lower bound $\frac{u}{2} - O(\log u)$ for $b \geq \frac{u}{2}(H_u - 1) + au$; since the independent check, $\frac{u}{2} - O_K(\log u)$ for $b \geq \frac{u}{2}(H_u - 1) - Ku$, which covers Part I's whole window at the saddle. (Row corrected 5 October 2026; it first read "with two-sided bounds for all (b, u) ", but the lower bound has a hypothesis.) Joint large-deviation principle for $(U_n \log N/N, H_n \log N/N)$, marginal rates I_r, L^p convergence to 1, $\mathbb{E}U_n \sim \mathbb{E}H_n \sim n/\log n$; deficit rate d/r	Part I, Theorem 1.1, (1.7), Section 7 Part II, Theorem 11.1 (second route: root-chain lower bound and the coarse majorant) Part I, Theorem 1.2, Section 8 Part II, Theorem 11.2 Part I, Section 2, Lemma 2.1; Part II, Section 12, Lemma 12.1 Part I, Lemma 3.1 (printed once; it is also the upper half of Part II's Theorem 13.1) Part I, Lemma 4.1, Proposition 4.2, Section 5 [write] Remark 18.1; [write] Proposition 18.2, after the independent check (Part II, Section 18) Part II, Theorem 11.3, Corollary 11.4

Open in both Parts

A multiplicative equivalent, an amplitude, a multiplicative transseries; the first scale of $\log A_s(n) - M_r(N)$ beyond all fixed orders (only $-O(\log N)$ and $O(R_r(N))$ are known); the typical size of $U_n - H_n$; fluctuation scales and limit laws of U_n and H_n ; the fraction of a longest unary path that is a consecutive root chain; uniform control of the weighted-placement sum; explicit constants. See Sections 9.2 and 20.

Finite computations in both Parts (exact recurrences and reduced-shape enumerations, coefficient series, corruption campaigns) are checks of formulas and implementations; no asymptotic statement rests on them. The literature comparisons recorded in the Parts are bounded and are not priority certificates; the write did not re-check them against the cited papers.

Notation across the two Parts

Common to both Parts, with the same meaning: $s, r = s + 1, N = n + 1, b$ (applications), u (abstractions), the admissibility condition ($N - u$ a positive multiple of r), $F_N(u) =$

$\frac{N-u}{r} \log(4u)$, C_b , $B_{m,k}$, $H_{m,k}$, $d_m = \sqrt{1 - m/u}$, $\rho = 1/(4u)$, $K = e(2 + \sqrt{3})$ and $c = \log K$, $\tau = 2 - \sqrt{3}$, the shape series D and its coefficients d_u , the reduced trees, W (the positive Lambert function) and the threshold inverse $\nu_s(y) = \min\{n : A_s(n) \geq y\}$. Report 109 writes $A_s(n, u)$ for Report 110's $A(b, u)$. The letters below change meaning between the Parts or inside one; each Part's own text fixes its meaning there, and each Part opens with a short table of its reading conventions.

Labels. Six label names occur in both manuscripts: `eq:B0`, `eq:G`, `eq:size`, `eq:uniform`, `sec:inverse`, `thm:inverse`. Under the prefixes `lam:ao:` and `lam:bg:` they are distinct; two of them name different mathematics:

Delivered name	Part I (Report 110)	Part II (Report 109)
<code>eq:G</code>	(6.1): the coarse envelope $G_N(u) = F_N(u) + cu - \log 2$	(16.1): the placement envelope $G_N(u) = F_N(u) + u \log(2eN/u)$
<code>eq:uniform</code>	(4.7): Proposition 4.2, the height-optimized bound	(13.1): Theorem 13.1, $C_b u^{b+1} \leq A \leq 2uK^u(4u)^b$
<code>eq:size</code>	(1.1): $n = u + rb + s$	(12.1): $n = u + b + s(b + 1)$, the same relation
<code>eq:B0</code>	(2.2)	Report 109's (3.3), the same identity; printed once
<code>thm:inverse</code> , <code>sec:inverse</code>	Theorem 1.2, Section 8	Theorem 11.2, Section 15: the weaker inverse

Symbols. The most dangerous is u_* , which names two different saddles.

Symbol	Meanings
u_*	Part I: $u_* = N/t$ with $t = W(4eNe^{-r/2})$, the maximizer of $H_N(u) = F_N(u) + u/2$ ((1.2), (5.8)); $H_N(u_*) = M_r(N)$. Part II: $u_* = N/w$ with $w = W(4eN)$, the maximizer of F_N alone (Section 14); $F_N(u_*) = \frac{N}{r}(w - 2 + \frac{1}{w})$. <i>Tempting false reading:</i> they are the same point. Since $w - t \rightarrow r/2$, Part II's u_* is smaller by about $\frac{r}{2}N/(\log N)^2$, and the two maximal values differ by $\frac{N}{2\log N}(1 + o(1))$: exactly the gap between the two counting theorems.
t, w	Part I: $t = W(4eNe^{-r/2})$, the shifted Lambert value; t_y in the inverse. Part II: $w = W(4eN)$ is its Lambert value, and t is the scaled abstraction count $u \log N/N$ (the first argument of $J_r(t, x)$, Sections 16–17); in Report 109's Section 3 (printed in Part I) t marked abstractions in $T_m(x, t)$ and was the variable of $D(t)$, written z here. <i>False reading:</i> Part II's t is near 1, Part I's t is near $\log N$.
T	Part I: the series $T_m(x, z)$ (2.1); the truncation T_J (Lemma 7.1, Section 8). Part II: $T_n = U_n \log N/N$, a random variable (11.7).
z	Part I: the abstraction marker in $T_m(x, z)$; $z = \ell - a$ (1.6); $z = 2 \log v - B$ in Section 8; the variable of the shape series $D(z)$. Part II: the size marker of $L_m(z)$ (12.5).
L, ℓ, k	Part I: $L = \log N$, $\ell = \log L$ (1.6); $L_u(x)$ the single-spine series (5.1); in the inverse $Y = \log y$, $v = \log Y$, $k = \log v$. Part II: $L = \log y$, $\ell = \log L$, $k = \log \ell$ in Theorem 11.2 and Section 15 (Part I's Y, v, k); $L_m(z)$ the size series (12.5).
K	$K = e(2 + \sqrt{3})$ in both. Part I also: $K_r = e^{r/2}/(4er)$, a constant (Theorem 1.2). Part II also: $K_r(d) = d/r$, a rate function (Corollary 11.4).
H	$H_{m,k}$ the normalized values (both). Part I: $H_N(u)$ the enhanced envelope (5.7); H_u the harmonic number (5.4). Part II: H_n the maximum unary height of a random term (11.7).
G_N	See the label table: two different envelopes.
$E, \mathbb{E}$	Part I: $E(b, u)$ the exponent of Proposition 4.2. $\mathbb{E}$ is an expectation in both.
J, I	Part I: J the negative-binomial sum (5.3); J a fixed truncation order; $I_N = [N/(2L), 2N/L]$. Part II: J_r the joint rate, I_r the marginal rate.

Symbol	Meanings
R	Part I: $R_r(N)$ the error scale (1.3). Part II: $R(b, u)$ the ratio (18.1).
Q	Part I: $Q_u(x)$ the spine product (5.1). Part II: $Q_h(u)$ an envelope in Section 17; in Lemma 3.2 (Report 109's, printed in Part I) Q , U , V , c count binary vertices, unary vertices, leaves and components of a forest.
D, d	$D(z)$ the shape series and d_u, d_m (both). Part I: $D = 1 + z $ (Lemma 7.1); $d = u - h$ the height deficit of a reduced tree (2.4). Part II: D_n the scaled deficit variable and d its value (Corollary 11.4); D a denominator and d a leaf depth in Sections 12 and 15.
h, q	Part I: h the largest unary depth of a terminal leaf of a reduced tree (for a term, Part II's H_n); q its number of binary vertices; q_j the inverse polynomials. Part II: $q = \log N$ and $h = \log q$ in Section 14; h a height value in Section 17.
a, c_0	Part I: $a = 1 + \log 4 - r/2$ (1.6); $a_0 = (\log 2 - \frac{1}{2})/2$, the Cantelli constant. Part II: $a_s(n, m)$ coefficients (12.6); $c_0 = \log 4 - 1$ and $a = \log r + c_0$ in Section 15; $0 < a < 1$ a threshold in Sections 16–17.
α, μ	Part I: $\alpha = b/u$, α_* ; $\mu = \mathbb{E}J$. Part II: α_i the slot occupation numbers in (12.3).
A, B, M	$A(b, u)$, $A_s(n)$ (both); Part I also $A = r/2 - 2$, $B = -\log K_r$ in Section 8, and A a constant in the proof of Proposition 4.2. Part II: $A_s(n, u, h)$; $\mathcal{B}_b$ the binary trees. M : Part I's $M_r(N)$; in Part II a bracketing constant (Section 15), a binomial argument (Section 16) and a size bound (Section 12).
x, v, n_0	x marks applications (both); Part II also $x = H_n \log N/N$'s value, and x the stationary variable of (16.3). v : Part I $v = \log Y$ or a vertex; Part II the number of leaves (12.1), $v = \log n_0$, or a vertex. n_0 (Part II's inverse approximant) corresponds to Part I's $N_0 - 1$.

Provenance and merge decisions

Two manuscripts of the session bundle of Reports 1–243, both dated 2 October 2026, with the author lines “Report 110” and “Report 109”; neither names an author or a tool, and neither carries “prepared for private review” wording. They arrived in commit 60f54ea06 and were placed by 9c995cefe (batch 103). Neither pins a ProveIt commit or names a repository path; Report 110 does not name Report 109, but its Section 9 says that it “does not repeat separate large-deviation results for binding geometry”, which are Report 109's.

- **Part I**, bundle Report 110, archive `Closed_Lambda_Terms_All_Logarithmic_Orders_Source.zip` (415,377 bytes, 24 files; `report110.tex`, 589 lines, 16 pages). The base: its `report110.tex` became this `article.tex`. Contributes the uniform logarithmic approximation with every fixed inverse-logarithmic order, the height-sensitive majorant, the single-spine lower bound, the two-stage localization, the polynomial recurrences and the controlled inverse.
- **Part II**, bundle Report 109, archive `Closed_Lambda_Terms_Growth_and_Binding_Geometry_Source.zip` (469,757 bytes, 20 files; `report109.tex`, 647 lines, 16 pages). Contributes the skeleton sandwich and exact weight formula, the size recurrence, the fuller write-up of the coarse majorant, the joint large-deviation principle for abstractions and unary height with its corollaries, the four-scale count and the first inverse (now a second route), the ratio frontier, and the table of initial terms.

Where the merge had to choose:

1. *Base and order*. Report 110 is the base and Part I, because its Theorems 1.1 and 1.2 imply Report 109's Theorems 1.1 and 1.2 with smaller errors under the same hypotheses (the precedent of the merged report a202059-ascent-100-110-growth, whose base, the later Report 243, sharpens the main estimates of its member, the earlier Report 241). Report 109 keeps sole credit for the random-geometry layer and the skeleton sandwich.

2. *One proof printed once.* Report 110’s Lemma 3.1 and the upper half of Report 109’s Theorem 3.1 are the same statement with the same proof (normalization at $\rho = 1/(4u)$, the depth forest, padding to $(u!)^2$, the shape series). The proof is printed once, in Part I, in Report 109’s fuller wording (its Sections 3.1–3.3, Lemmas 3.2–3.3 and Remark 3.4); Report 110’s condensed proof follows the same steps in the same order and is not reprinted (it is in the arrival commit). In that moved text Report 109’s variable t of the shape series is written z , as in Report 110, because t is Part I’s Lambert value; this is the only renamed symbol. Report 109’s statement of its Theorem 3.1 stays in Part II. Report 109’s radical-recurrence identities (its (3.2)–(3.5)) are Report 110’s (2.1)–(2.3), printed once in Part I.
3. *Same theorem, different place.* Report 110’s Lemma 2.1 is the reduced-shape formula of Report 109’s checks section (Section 19), with the same stars-and-bars proof; both are printed, the second with a note, because Report 109 states it as a finite check.
4. *Second routes.* Report 109’s Theorems 1.1 and 1.2 are printed in full with their proofs (Sections 14 and 15) as a weaker second route, with dated notes; the statements of Report 109 that Report 110 makes stale (the undetermined coefficient of $n/\log n$, the residual factor $\exp\{O(n/\log n)\}$, the undetermined weighted-placement enhancement, the constant over ℓ in the inverse) are printed as delivered with dated notes.
5. *Questions.* Report 110’s Question 2 and Report 109’s Question 2 ask the same thing (the typical off-spine deficit $u - h$, that is $U_n - H_n$); they are merged into item 2 of Section 20. Report 110’s Question 3 and Report 109’s Question 4 are merged into item 3 of Section 9.2.
6. *Front matter and bibliography.* The two title blocks and abstracts are printed at the head of their Parts; one table of contents replaces two. The two bibliographies are merged into one, keeping every detail of every entry; three works appear in both under different keys (Report 109’s `bci`, `david`, `restricted` are Report 110’s `bggj2013`, `david2013`, `bggj2018`), and Report 110’s keys are used.
7. *Write additions.* Remark 18.1 with its proof, Proposition 18.2 with its proof and the record of the independent check of the write (both added after that check, 5 October 2026), the two “Further questions and research” sections, the dated notes, the reading-conventions tables and this front matter. Everything else is delivered text.

Relation to neighbouring reports and to the formal project

No other report of the repository counts lambda terms: at the placement, a search of every tracked text file (`lib/` excluded) found no mention of A135501 or A220894, and the Bodini and Gittenberger papers cited elsewhere in the collection (for example in `a082161-air-y-amplitudes`) concern other objects. The reports `a088714-bell-scale-growth` and `a277364-bell-asymptotics` of the same directory also solve Lambert- W saddle equations; they share no theorem with this one. The threshold inverse $\nu_s(y)$ is the integer staircase of Definition `p0:def:three-inverses` of the transseries volume (`Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`); both Parts bracket it by monotonicity from bounds on the logarithm, and neither uses that volume’s staircase theorem, which needs an admissible interpolation.

The set counted here is a type of the repository’s formal combinatory-logic project. `Computability/CombinatoryLogic/Lean/CombinatoryLogic/Lambda.lean` (line 15) defines inductive `Term : Nat → Type` with constructors `var (index : Fin n)`, `app`

and `lam (body : Term (n + 1))`, intrinsically scoped de Bruijn terms; `Term 0` is exactly the set of closed alpha-classes counted by both Parts, and its constructor recursion is the binder-context recursion (2.1) (with Part II's binder positions, Section 12). The Rocq development has the equivalent `term V` with `lam : term (option V) -> term V` (`Computability/CombinatoryLogic/Coq/Lambda.v`, line 16). Neither development defines a size or counts terms, and no statement of this report is formalized; placement beside that project confers no formal status.

Part I

All logarithmic orders and a controlled threshold inverse

Bundle Report 110, archive `Closed_Lambda_Terms_All_Logarithmic_Orders_Source.zip`, delivered as `report110.tex` (16 pages), dated 2 October 2026. Delivered title “All logarithmic orders for closed lambda terms: Uniform counting bounds and a controlled threshold inverse”; author line “Report 110”. Printed as delivered, with [write] notes; section, statement and equation numbers are those of the delivery, except in Section 3 (see the Guide). Section 9.2 is added in the write.

Abstract of this Part (as delivered). We determine every fixed inverse-logarithmic order in the logarithms of two unrestricted closed lambda-term counts: OEIS A135501 and A220894. If variable occurrences have size $s \in \{0, 1\}$, while abstractions and applications have size one, put $r = s + 1$, $N = n + 1$, and $t = W(4eNe^{-r/2})$. Then

$$M_r(N) - O(\log N) \leq \log A_s(n) \leq M_r(N) + O\left(N^{1-1/(3r)}(\log N)^{-2/3+1/(3r)}\right),$$

where $M_r(N) = \frac{N}{r}(t - 2 + t^{-1}) + \frac{N}{2}$. A height-sensitive reduced-tree majorant supplies the upper bound. An exact single-spine generating function, Catalan convolution, and a one-sided variance bound supply the lower bound at the moving saddle. Two localization stages make the error uniform. We give arbitrary-order polynomial recurrences with analytic remainders and a controlled inverse, including its asymmetric threshold bracket. The report is self-contained and does not assume that a fixed-abstraction singular expansion is uniform. The conclusions concern logarithms; they do not establish a multiplicative equivalent, an amplitude, or a full multiplicative transseries. Reproducible finite checks are supplementary to the analytic proofs.

Reading conventions. This Part’s letters are the reference letters of the report. In particular $t = W(4eNe^{-r/2})$ is a Lambert value near $\log N$, $u_* = N/t$, z marks abstractions in $T_m(x, z)$, and $L = \log N$; Part II uses t , u_* , L otherwise (front matter, “Notation across the two Parts”).

1 Models and main conclusions

A lambda term is built from variables, abstractions $\lambda x.M$, and ordered applications (MN) . Alpha-equivalence identifies consistent changes of bound-variable names. A term is closed if every variable occurrence is bound by an enclosing abstraction. The two sequences considered here count all closed terms up to alpha-equivalence, without typing, normalization, linearity, or index restrictions.

Let variables have size $s \in \{0, 1\}$ and abstractions and applications size one. Write $A_s(n)$ for the number of closed alpha-classes of size n . Then

$$A_1(n) = \text{A135501}(n), \quad A_0(n) = \text{A220894}(n).$$

The index starts at $n = 0$. The term $\lambda x.x$ has size two for $s = 1$ and size one for $s = 0$. All logarithms are natural. If a skeleton has b application vertices and u abstraction vertices, it has $b + 1$ variable leaves. Consequently, with $r = s + 1$ and $N = n + 1$,

$$n = u + rb + s, \quad b = \frac{N - u}{r} - 1. \tag{1.1}$$

An admissible u is a positive integer with $N - u$ a positive multiple of r . This congruence is retained throughout. Let $A(b, u)$ count the closed terms with these fixed parameters; the size convention does not otherwise affect it.

The positive Lambert function satisfies $W(z)e^{W(z)} = z$ for $z > 0$. Define

$$t = W(4eNe^{-r/2}), \quad u_* = N/t, \quad (1.2)$$

$$M_r(N) = \frac{N}{r} \left(t - 2 + \frac{1}{t} \right) + \frac{N}{2}, \quad R_r(N) = N^{1-1/(3r)} (\log N)^{-2/3+1/(3r)}. \quad (1.3)$$

Theorem 1.1 (Uniform logarithmic approximation). *For each $s \in \{0, 1\}$ and $n \rightarrow \infty$,*

$$M_r(N) - O(\log N) \leq \log A_s(n) \leq M_r(N) + O(R_r(N)). \quad (1.4)$$

The constants depend at most on r . Since there are only two choices, common constants can be chosen.

For every fixed real k ,

$$\frac{R_r(N)}{N/(\log N)^k} = N^{-1/(3r)} (\log N)^{k-2/3+1/(3r)} \longrightarrow 0. \quad (1.5)$$

Thus the main expression determines every fixed inverse-logarithmic order of the log-count. This is stronger than merely expanding a candidate saddle expression formally: the error in the full count lies below every scale being identified.

[write] **5 October 2026.** Part II, Theorem 11.1, is a weaker form of (1.4): $\log A_s(n) = \frac{N}{r}(w - 2 + \frac{1}{w}) + O(N/\log N)$ with $w = W(4eN)$, proved by a second route (the root-chain lower bound $C_b u^{b+1}$ of Lemma 12.1 and the coarse majorant of Lemma 3.1). Theorem 1.1 implies it: by the expansion of Section 7 with $J = 1$ and Part II's expansion of $w - 2 + 1/w$ (Section 14), $M_r(N) - \frac{N}{r}(w - 2 + \frac{1}{w}) = \frac{N}{2 \log N} + O(N(\log \log N)^2/(\log N)^2)$, which lies inside Part II's error. The upper error $R_r(N)$ is an upper bound only: its exponent $1 - 1/(3r)$ comes from maximizing a majorant over height deficits (Proposition 4.2, (6.6)), and nothing here shows that $\log A_s(n) - M_r(N)$ is ever of that size; this Part says so itself in Section 9 (question 1). The true residual lies between $-O(\log N)$ and $O(R_r(N))$.

For an explicit form, set

$$L = \log N, \quad \ell = \log L, \quad a = 1 + \log 4 - r/2, \quad z = \ell - a. \quad (1.6)$$

Section 7 proves that for every fixed integer $J \geq 1$,

$$\log A_s(n) = \frac{N}{r} \left[L - \ell + \log 4 - 1 + \sum_{j=1}^J \frac{P_j(z)}{L^j} \right] + O\left(\frac{N(1 + \ell)^{J+1}}{L^{J+1}} \right). \quad (1.7)$$

In particular, the coefficient at the newly separated $n/\log n$ scale is determined by

$$\frac{n}{r} \frac{\log \log n - \log 4 + r/2}{\log n}.$$

The polynomial recurrences specify arbitrary fixed order; they make no assertion about truncation orders growing with n .

[write] **5 October 2026.** This is the coefficient that Part II says its own theorem does not determine (the sentence after Theorem 11.1). Its constant part is $\frac{r/2 - \log 4}{r} = \frac{1}{2} - \frac{\log 4}{r}$: $\frac{1}{2} - \log 2 \approx -0.193$ for A135501 ($r = 2$) and $\frac{1}{2} - \log 4 \approx -0.886$ for A220894 ($r = 1$). Part II's Lambert expression has $-\frac{\log 4}{r}$ there; the difference $\frac{1}{2} \cdot \frac{n}{\log n}$ is the factor $e^{u/2+o(u)}$ by which the full count exceeds the root-chain subclass at the saddle (Remark 18.1).

1.1 The threshold inverse

Define

$$\nu_s(y) = \min\{n \in \mathbb{N} : A_s(n) \geq y\}, \quad Y = \log y.$$

Adding an unused root abstraction is an injection from size n into size $n + 1$ in both models, so $A_s(n)$ is nondecreasing. Theorem 1.1 makes it unbounded.

Theorem 1.2 (Controlled inverse). *Let N_0 be the unique sufficiently large solution of $M_r(N_0) = Y$. Then*

$$\nu_s(y) = N_0 - 1 + O\left(\frac{R_r(N_0)}{\log N_0} + 1\right). \quad (1.8)$$

More precisely,

$$N_0 - O\left(\frac{R_r(N_0)}{\log N_0}\right) \leq \nu_s(y) + 1 \leq N_0 + O(1). \quad (1.9)$$

Put

$$K_r = \frac{e^{r/2}}{4er}, \quad \Pi_r(t) = t^2 + (r/2 - 2)t + 1.$$

If t_y is the unique large positive solution of

$$Y = K_r e^{t_y} \Pi_r(t_y), \quad \text{then} \quad N_0 = \frac{rY t_y}{\Pi_r(t_y)}. \quad (1.10)$$

In the variables $v = \log Y$ and $k = \log v$,

$$\nu_s(y) = \frac{rY}{v - 2k + \log(4r) - 1 + [4k + 1 + r/2 - 2\log(4r)]/v} + O\left(\frac{Y(1+k)^2}{v^4}\right). \quad (1.11)$$

An arbitrary-order inverse recurrence with a proved remainder is given in Section 8.

The notation Π_r distinguishes the exact quadratic from the forward coefficient polynomials P_j . The error terms in these statements are asymptotic, with no claimed explicit numerical onset.

[write] **5 October 2026.** Part II, Theorem 11.2, is the weaker form of (1.11). In this Part's letters ($Y = \log y$, $v = \log Y$, $k = \log v$; Part II writes L , ℓ , $\log \ell$) it reads $\nu_s(y) = rY/(v - 2k + \log r + \log 4 - 1 + 4k/v) + O(Y/v^3)$: the first four terms of the denominator coincide with those of (1.11), and this Part adds the term $[1 + r/2 - 2\log(4r)]/v$, whose coefficient is ≈ -1.273 for $r = 1$ and ≈ -2.159 for $r = 2$. Because that coefficient is not zero, the two approximations differ by $r|1 + r/2 - 2\log(4r)|Y/v^3(1 + o(1))$, so Part II's error $O(Y/v^3)$ is of exact order at its truncation, and its approximation lies below $\nu_s(y)$ for large y . The asymmetric bracket (1.9) and the inverse recurrence of Section 8 have no counterpart in Part II.

2 Exact recurrence and reduced trees

Let $T_m(x, z)$ count terms with m available outer binders, where x marks applications and z marks abstractions. A variable occurrence chooses one of its available binders. This is the alpha-class encoding, so no labeling factorial is present. The root decomposition is

$$T_m = m + zT_{m+1} + xT_m^2. \quad (2.1)$$

Write $B_{m,k}(x) = [z^k]T_m(x, z)$. Then

$$B_{m,0}(x) = \frac{1 - \sqrt{1 - 4mx}}{2x}, \quad (2.2)$$

$$B_{m,k}(x) = \frac{B_{m+1,k-1}(x) + x \sum_{j=1}^{k-1} B_{m,j}(x) B_{m,k-j}(x)}{\sqrt{1 - 4mx}}, \quad k \geq 1. \quad (2.3)$$

The value at $x = 0$ in (2.2) is interpreted as m . To obtain (2.3), move the two terms $x B_{m,0} B_{m,k}$ to the left and use $1 - 2x B_{m,0} = \sqrt{1 - 4mx}$. These are identities of nonnegative formal power series, and

$$A(b, u) = [x^b] B_{0,u}(x).$$

The recursion in (2.3) terminates: a unary branch decreases k by one, and both children of a binary branch have smaller positive unary counts.

[write] 5 October 2026. Part II derives the same series in its Section 3 (Report 109's (3.2)–(3.5)) with t instead of z marking abstractions; its identities are (2.1)–(2.3) and the display above, printed once here. Part II, Section 12, adds the binder-position reading of a fixed skeleton (a term is a skeleton with a choice of ancestor abstraction at each leaf), the size recurrence (12.6) and the skeleton sandwich, which this Part does not use.

Expanding this finite recursion gives plane reduced trees with exactly u unary vertices. A unary vertex increments m ; a binary vertex has two children each containing at least one unary vertex. A terminal leaf carries $B_{m,0}$. Every terminal leaf therefore has a unary parent. At a vertex, m is the number of strict unary ancestors. Every internal vertex contributes a radical $(1 - 4mx)^{-1/2}$ and every binary vertex also contributes x .

Let q be the number of binary vertices and let h be the largest unary depth of a terminal leaf. Write $d = u - h$. A deepest path contains h unary vertices. There are $q + 1$ terminal leaves in all. The other q leaves have distinct unary parents off this path, because a unary vertex on the selected path has its unique child on that path. Hence

$$1 \leq h \leq u, \quad q \leq d = u - h. \quad (2.4)$$

Every terminal depth is at most h . Every internal depth is at most $h - 1$, since its descendant subtree still contains a unary vertex.

Lemma 2.1 (Exact reduced shape count). *For $u \geq 1$ and $0 \leq q \leq u - 1$, the number of reduced shapes with u unary and q binary vertices is*

$$C_q \binom{u + q - 1}{2q}, \quad C_q = \frac{1}{q + 1} \binom{2q}{q}. \quad (2.5)$$

Proof. Suppress all unary vertices. The remaining full plane binary tree has q binary vertices and is counted by C_q . Its $2q + 1$ incoming vertex slots, including the root slot, receive unary chains. Every one of the $q + 1$ leaf slots must contain at least one unary. After assigning these mandatory vertices, distribute $u - q - 1$ remaining unaries among $2q + 1$ slots. Stars and bars gives the binomial in (2.5). Conversely every such assignment is a permitted reduced tree. For $q = 0$ this gives the unique unary chain. $\square$

[write] 5 October 2026. The same formula, with the same contraction of unary chains and the same stars-and-bars count, is stated and proved in Part II's checks section (Section 19, item 7 and the paragraph after the list), where its weighted sum $d_u = \sum_q 2^{-q} C_q \binom{u+q-1}{2q}$ is compared through $u = 100$ with the quadratic shape recurrence.

3 A coarse bound for global localization

We include the uniform majorant needed to exclude the tails of the abstraction count. This makes the later moving-saddle argument independent of an external asymptotic result.

Lemma 3.1 (Coarse uniform majorant). *For every integer $u \geq 1$ and $b \geq 0$,*

$$A(b, u) \leq 2u K^u (4u)^b, \quad K = e(2 + \sqrt{3}). \quad (3.1)$$

[write] **5 October 2026.** Proved in both manuscripts by one argument: by Report 110 in a condensed proof, and by Report 109 as the upper half of its Theorem 3.1 (Theorem 13.1 in Part II), with its Lemmas 3.2–3.3 and its Section 3.3. The proof is printed once, here, in Report 109’s fuller wording; it occupies Sections 3.1–3.3 below, and Lemmas 3.2 and 3.3 and the remark at the end of Section 3.3 keep Report 109’s numbers. Report 110’s condensed proof takes the same steps in the same order (finiteness at $\rho = 1/(4u)$ by induction on k , the normalization by $2u$, the bound $2(u - m) - 1$ for the internal vertices of unary depth at least m , padding the deficit list to $(u!)^2$, and the shape series with $D(\tau) = \sqrt{3} - 1$); it is not reprinted and remains in the arrival commit. Report 109’s derivation of (2.2)–(2.3) is omitted as a repetition of Section 2; its references to its own (3.4) point to (2.3), and its shape variable t is written z , as in Report 110.

3.1 The radical recurrence and its endpoint

Fix $u \geq 1$ and set $\rho = 1/(4u)$. Every member of the finite triangular family $B_{m,k}(\rho)$ with $m + k \leq u$ is finite. Indeed, the Catalan values $B_{m,0}(\rho)$ are finite even at $m = u$, where the square root vanishes but its inverse is not used. If $k \geq 1$, then $m < u$, so the denominator in (2.3) is strictly positive. Induction on k proves finiteness of the numerator and the endpoint value. Nonnegative coefficients identify the limit as $x \uparrow \rho$ with the sum of the series at ρ . Thus this is a valid positive-series endpoint evaluation, not a singularity-transfer approximation.

Normalize by

$$H_{m,k} = \frac{B_{m,k}(\rho)}{2u}, \quad d_m = \sqrt{1 - m/u}.$$

Then

$$H_{m,0} = 1 - d_m \in [0, 1], \tag{3.2}$$

$$H_{m,k} = d_m^{-1} \left(H_{m+1,k-1} + \frac{1}{2} \sum_{j=1}^{k-1} H_{m,j} H_{m,k-j} \right). \tag{3.3}$$

The coefficient $1/2$ is exactly $\rho(2u)$.

3.2 Reduced trees and a depth forest

Expand (3.3) from $(m, k) = (0, u)$. A unary expansion consumes one abstraction and increments m ; a binary expansion splits a positive remaining unary count into two positive parts. A terminal is reached only by a unary expansion. These expansion histories are finite plane unary–binary trees with exactly u unary vertices, and every terminal leaf has a unary parent. They are *reduced* trees: their binary count q need not be the original application count b .

The subtree unary counts uniquely recover every convolution index, so no additional multiplicity factor is missing. If $m(v)$ counts the strict unary ancestors of an internal vertex v , a reduced shape has weight

$$2^{-q} \prod_{v \text{ internal}} (1 - m(v)/u)^{-1/2} \prod_{v \text{ terminal}} H_{m(v),0}. \tag{3.4}$$

Each terminal factor is at most one. There are $q + 1$ terminal leaves, all with distinct unary parents, so $q \leq u - 1$ and there are at most $2u - 1$ internal vertices.

Lemma 3.2 (Depth-forest inequality). *If a reduced tree with u unary vertices has any internal vertex of unary depth at least m , the number of such internal vertices is at most*

$$2(u - m) - 1. \tag{3.5}$$

Proof. Take all internal vertices at unary depth at least m , together with their descendant terminal leaves. They form a descendant-closed forest with $c \geq 1$ components. At least m unary vertices lie outside this forest: follow a path to any component root and take its first m unary ancestors. Thus the forest has $U \leq u - m$ unary vertices. If it has Q binary vertices and V leaves, the forest identity is $V = Q + c$. Every included terminal still has its unary parent in the forest, hence $V \leq U$. It follows that

$$U + Q \leq 2U - c \leq 2(u - m) - 1.$$

A terminal whose parent has depth $m - 1$ is not an isolated component of this forest: it is not a descendant of any included internal vertex. This convention is needed for $V \leq U$. $\square$

Lemma 3.3 (Uniform radical product). *For every reduced shape with u unary vertices,*

$$\prod_{v \text{ internal}} (1 - m(v)/u)^{-1/2} \leq \frac{u^{u-1/2}}{u!} \leq e^u. \quad (3.6)$$

Proof. Let $\delta(v) = u - m(v) \in \{1, \dots, u\}$. Applying Lemma 3.2 at depth $u - j$, at most $2j - 1$ deficits are at most j . Pad the deficit list with copies of u until its length is $2u - 1$. The sorted padded list is coordinatewise at least

$$1, 2, 2, 3, 3, \dots, u, u,$$

so its product is at least $(u!)^2$. Padding preserves the squared radical ratio, since each extra entry contributes $u/u = 1$. The ratio is therefore at most $u^{2u-1}/(u!)^2$. Finally, $u! \geq (u/e)^u$ gives the claimed bound, including $u = 1$. $\square$

3.3 The weighted shape generating function

Let d_u be the sum of 2^{-q} over reduced shapes with u unary vertices. A root is either unary with a terminal or nonempty child, or binary with two nonempty children. Thus

$$D(z) = \sum_{u \geq 1} d_u z^u = z(1 + D(z)) + \frac{1}{2} D(z)^2, \quad D(z) = 1 - z - \sqrt{1 - 4z + z^2}. \quad (3.7)$$

At $\tau = 2 - \sqrt{3}$, monotone convergence of the nonnegative series gives

$$D(\tau) = \sqrt{3} - 1 < 1, \quad d_u \leq \tau^{-u} = (2 + \sqrt{3})^u.$$

Combining this with (3.4) and Lemma 3.3 proves

$$H_{0,u} \leq [e(2 + \sqrt{3})]^u = K^u, \quad B_{0,u}(\rho) \leq 2uK^u.$$

Positivity and endpoint finiteness now give

$$[x^b]B_{0,u}(x) \leq \rho^{-b}B_{0,u}(\rho) \leq 2uK^u(4u)^b.$$

The lower bound in Theorem 13.1 is Lemma 12.1, completing the proof.

[write] **5 October 2026.** The last display is (3.1), which proves Lemma 3.1. The final sentence concerns Report 109's two-sided statement, Theorem 13.1.

Remark 3.4 (Boundary sanity checks). When $b = 0$, the exact count is u ; when $u = 1$, it is C_b . In the endpoint evaluation, $m = u$ occurs only at a terminal with $k = 0$, so there is no division by zero. Closed terms with $u = 0$ do not occur and are never assigned a logarithm.

4 A height sensitive uniform upper bound

The key improvement comes from retaining the height before coefficient extraction. Let $A_h(b, u)$ count the terms belonging to reduced shapes with maximum unary depth h .

Lemma 4.1 (Height bound). *For $b \geq 1$, $1 \leq h < u$, and $d = u - h$,*

$$A_h(b, u) \leq 2he^{h/2}(4h)^b(d+1) \left[\frac{2e^3 u^2(b+d)}{d^3} \right]^d. \quad (4.1)$$

For $h = u$,

$$A_u(b, u) \leq 2ue^{u/2}(4u)^b. \quad (4.2)$$

Proof. Fix the height subclass first. For $0 < \delta < 1$ evaluate only its finite reduced-tree products at

$$x = \frac{1 - \delta}{4h}.$$

Every such product converges there: terminal parameters satisfy $m \leq h$ and internal parameters satisfy $m < h$. When $h < u$, this radius may exceed the unrestricted radius $1/(4u)$; we never evaluate the unrestricted $B_{0,u}$ there.

Normalize every factor B in (2.3) by $2h$. A binary vertex has normalized factor $2hx = (1 - \delta)/2 \leq 1/2$. For $0 \leq m \leq h$, a terminal factor satisfies

$$\frac{B_{m,0}(x)}{2h} = \frac{1 - \sqrt{1 - (1 - \delta)m/h}}{1 - \delta} \leq \frac{1 - \sqrt{\delta}}{1 - \delta} = \frac{1}{1 + \sqrt{\delta}} \leq 1. \quad (4.3)$$

There is no extra factor $(1 - \delta)^{-1}$ to retain. The $q + 1$ terminal normalizations and q binary normalizations leave one exterior factor $2h$.

On a selected deepest path, the h unary vertices have depths $0, 1, \dots, h - 1$. Their radical product is at most

$$\prod_{m=0}^{h-1} (1 - m/h)^{-1/2} = \sqrt{h^h/h!} \leq e^{h/2}.$$

There are $u + q - h = d + q \leq 2d$ other internal vertices, each with radical at most $\delta^{-1/2}$. A normalized shape contribution is therefore at most

$$e^{h/2} \delta^{-d} 2^{-q}. \quad (4.4)$$

For $h = 1$, the distinguished unary radical is simply one, so the argument still applies.

For $1 \leq q \leq d \leq u - 1$, the elementary estimates $C_q \leq 4^q$ and $\binom{n}{k} \leq (en/k)^k$ give

$$2^{-q} C_q \binom{u + q - 1}{2q} \leq [2e^2(u/q)^2]^q.$$

The function $v \log(2e^2(u/v)^2)$ increases on $0 < v \leq u$, since its derivative is $\log 2 + 2 \log(u/v) > 0$. Including the $q = 0$ summand gives

$$\sum_{q=0}^d 2^{-q} C_q \binom{u + q - 1}{2q} \leq (d + 1) [2e^2(u/d)^2]^d. \quad (4.5)$$

This overcounts heights other than h , which is harmless; every desired shape has $q \leq d$. Positivity and (4.4)–(4.5) imply

$$A_h(b, u) \leq 2he^{h/2}(4h)^b(d+1)(1 - \delta)^{-b} \delta^{-d} [2e^2(u/d)^2]^d.$$

Choose $\delta = d/(b+d)$. Then $(1-\delta)^{-b} = (1+d/b)^b \leq e^d$, proving (4.1).

If $d = 0$, (2.4) forces $q = 0$: there is one reduced unary chain. Evaluate it at $x = 1/(4u)$. Its terminal factor is $2u$ and its radical product is at most $e^{u/2}$. All internal depths are below u , so this endpoint is legitimate. This proves (4.2), also for $u = 1$. $\square$

Proposition 4.2 (Uniform fixed parameter estimate). *For $u \geq 1$, $b \geq 0$, and $\alpha = b/u$, put*

$$E(b, u) = 3 \cdot 2^{1/3} u (1 + \alpha)^{1/3} e^{-\alpha/3}. \quad (4.6)$$

Then

$$A(b, u) \leq 2u^2(u+1)(4u)^b \exp\{u/2 + E(b, u)\}, \quad (4.7)$$

$$\log A(b, u) \leq b \log(4u) + u/2 + E(b, u) + O(\log(u+1)), \quad (4.8)$$

with an absolute uniform constant.

Proof. Relative to $(4u)^b e^{u/2}$, the exponential correction in (4.1) is at most

$$d \left[-\alpha + \log(2e^3(1+\alpha)) + 3 \log(u/d) \right].$$

Here we used $(h/u)^b \leq e^{-bd/u}$ and $b+d \leq u(1+\alpha)$, and discarded the favorable term $-d/2$. For $A = -\alpha + \log(2e^3(1+\alpha))$, the maximum of $v(A - 3 \log v)$ over all $v > 0$ is $3e^{A/3-1}$. Apply this with $v = d/u$ to get exactly $E(b, u)$. Summing at most u heights and using $2h \leq 2u$, $d+1 \leq u+1$ proves (4.7) when $b \geq 1$. The $d = 0$ term obeys the same bound. When $b = 0$, the only skeleton is a unary chain, with u choices for its final variable; thus $A(0, u) = u$ and the inequality is immediate. Finally $\log(2u^2(u+1)) \leq 4 \log(u+1)$. $\square$

[write] **5 October 2026.** Proposition 4.2 and the single-spine inequality (5.6) of the next section are the two halves of Remark 18.1 in Part II, which determines Part II's ratio $R(b, u) = A(b, u)/(C_b u^{b+1})$ to leading order at the moving saddle.

5 A single spine lower bound

A single-spine term is one in which every abstraction lies on one ancestor path. Applications and unary-free Catalan branches are allowed along that path; the abstractions need not form a consecutive initial chain. Iterating only the unary term of (2.3) gives its exact application generating function

$$L_u(x) = B_{u,0}(x) Q_u(x), \quad Q_u(x) = \prod_{m=0}^{u-1} (1 - 4mx)^{-1/2}. \quad (5.1)$$

Equivalently, between consecutive abstractions, a path application has two possible continuing children and a unary-free side branch. The context sequence has generating function $1/(1 - 2xB_{m,0}(x)) = (1 - 4mx)^{-1/2}$. This proves the interpretation without any singular asymptotic approximation.

At $\rho = 1/(4u)$ all factors of Q_u are finite, and

$$Q_u(\rho) = \sqrt{u^u/u!}, \quad \log Q_u(\rho) = u/2 + O(\log(u+1)). \quad (5.2)$$

The logarithmic factorial estimate follows from upper and lower integral bounds on $\sum_{j=1}^u \log j$.

Normalize these nonnegative coefficients into a probability law:

$$\mathbb{P}(J = j) = \frac{[x^j] Q_u(x) \rho^j}{Q_u(\rho)}, \quad \mathbb{E}[z^J] = \prod_{m=0}^{u-1} \left(\frac{1 - m/u}{1 - (m/u)z} \right)^{1/2}. \quad (5.3)$$

Each factor is a negative-binomial probability generating function of shape $1/2$; this displayed formula specifies the parameter convention. The factor with $m = 0$ is identically one. Thus J is a sum of independent such variables. Differentiating the factors gives

$$\mu = \mathbb{E}J = \frac{1}{2} \sum_{m=0}^{u-1} \frac{m}{u-m} = \frac{u}{2}(H_u - 1) \leq \frac{u}{2} \log u, \quad (5.4)$$

$$\sigma^2 = \text{Var } J = \frac{1}{2} \sum_{k=1}^{u-1} \frac{u(u-k)}{k^2} \leq u^2. \quad (5.5)$$

Here $H_u = \sum_{j=1}^u 1/j \leq 1 + \log u$, and $\sum_{j \geq 1} j^{-2} < 2$ follows by an integral bound. Empty sums for $u = 1$ cause no problem.

The Catalan ratio $C_{k+1}/(4C_k) = (2k+1)/(2k+4) < 1$ implies $C_{b-j} \geq 4^{-j}C_b$ for $0 \leq j \leq b$. Coefficient convolution in (5.1) therefore yields the exact lower inequality

$$A(b, u) \geq [x^b]L_u(x) \geq C_b u^{b+1} Q_u(\rho) \mathbb{P}(J \leq b). \quad (5.6)$$

Indeed the convolution sum is $\sum_{j=0}^b [x^j]Q_u C_{b-j} u^{b-j+1}$, and the stated Catalan inequality produces the factor ρ^j .

Define the two smooth functions

$$F_N(u) = \frac{N-u}{r} \log(4u), \quad H_N(u) = F_N(u) + u/2. \quad (5.7)$$

Direct differentiation gives

$$H'_N(u) = \frac{N/u - 1 - \log(4u)}{r} + \frac{1}{2}, \quad H''_N(u) = -\frac{1}{r} \left(\frac{N}{u^2} + \frac{1}{u} \right) < 0.$$

The derivative changes from positive to negative, so the maximum is unique. Its critical point is $u_* = N/t$, where

$$t + \log t = \log(4eN) - r/2, \quad \log(4u_*) = t - 1 + r/2, \quad H_N(u_*) = M_r(N). \quad (5.8)$$

In particular $t \sim \log N$ and $u_* \sim N/\log N$.

Choose an admissible integer u_0 within $r/2$ of u_* . For large N it lies in the admissible interior, since $u_* \rightarrow \infty$ and $N - r - u_* \rightarrow \infty$. Put $b_0 = (N - u_0)/r - 1$. The bounded rounding shift changes N/u by $O((\log N)^2/N)$, and

$$\frac{b_0}{u_0} = \frac{\log(4u_0)}{r} - \frac{1}{2} + o(1). \quad (5.9)$$

Combining (5.4) and (5.9),

$$\frac{b_0 - \mu}{u_0} \geq \left(\frac{1}{r} - \frac{1}{2} \right) \log u_0 + \frac{\log 4}{r} - \frac{1}{2} + o(1).$$

For $r = 2$, the right side tends to $\log 2 - 1/2 > 0$; for $r = 1$, it tends to infinity. Hence the common constant $a_0 = (\log 2 - 1/2)/2 > 0$ satisfies $b_0 - \mu \geq a_0 u_0$ eventually. Cantelli's inequality and (5.5) give

$$\mathbb{P}(J \leq b_0) \geq \frac{a_0^2}{1 + a_0^2} > 0. \quad (5.10)$$

For completeness, if X has mean zero and variance σ^2 , Markov's inequality applied to $(X+h)^2$ on $X \geq c > 0$ gives $\mathbb{P}(X \geq c) \leq (\sigma^2 + h^2)/(c+h)^2$. Taking $h = \sigma^2/c$ proves Cantelli; zero variance is immediate. Here $\{J > b_0\} \subseteq \{J - \mu \geq a_0 u_0\}$, so the integer endpoint is harmless.

The elementary bound $C_b \geq 4^b/((2b+1)(b+1))$, (5.2), (5.6), and (5.10) now imply

$$\log A_s(n) \geq H_N(u_0) - O(\log N).$$

Taylor's theorem at u_* loses only $O((\log N)^2/N) = o(1)$, because $|u_0 - u_*| \leq r/2$ and $|H_N''| = O((\log N)^2/N)$ on the intervening segment. This proves the lower half of Theorem 1.1, including both residue classes in the $r = 2$ model.

[write] 5 October 2026. Part II's lower bound is the root-chain subclass alone (all abstractions in the root slot, Lemma 12.1), which gives $C_b u^{b+1}$ and the envelope F_N . The single-spine class gains at least the factor $Q_u(\rho)\mathbb{P}(J \leq b) = e^{u/2+O(\log u)}$ near the saddle, by (5.6), (5.2) and (5.10). This factor is the term $u/2$ of the envelope $H_N(u) = F_N(u) + u/2$ of (5.7); at the saddle it is worth about $\frac{N}{2 \log N}$, the difference between $M_r(N)$ and Part II's Lambert expression.

6 Two stages of saddle localization

We prove the upper half without assuming a uniform fixed- u singular expansion. Put $L = \log N$ and $c = \log K$, where K is from Lemma 3.1. Since $b+1 = (N-u)/r$, the coarse bound gives exactly

$$\log A(b, u) \leq G_N(u) := F_N(u) + cu - \log 2. \quad (6.1)$$

There is no omitted u -dependent prefactor. This envelope is strictly concave, with maximizer $v_N = N/W(4eNe^{-rc}) \sim N/L$. For large N , this point lies in

$$I_N = [N/(2L), 2N/L].$$

Concavity implies that the largest envelope value outside I_N is bounded by one of its endpoint values. For fixed $x \in \{1/2, 2\}$,

$$\begin{aligned} G_N(xN/L) &= \frac{N}{r}[L - \log L + \log 4 + \log x - x] + o(N), \\ M_r(N) &= \frac{N}{r}[L - \log L + \log 4 - 1] + o(N). \end{aligned}$$

Both numbers $\log x - x + 1$ are strictly negative. There is therefore $\eta > 0$ such that every admissible $u \notin I_N$ satisfies

$$\log A(b, u) \leq M_r(N) - \eta N. \quad (6.2)$$

This includes the possible endpoint $b = 0$.

[write] 5 October 2026. The envelope G_N of (6.1) is not Part II's placement envelope G_N of (16.1).

Within I_N , $\alpha = (N/u - 1)/r - 1/u$ satisfies $L/(2r) - O(1) \leq \alpha \leq 2L/r$. Use these separately in the factors of (4.6) to obtain the uniform bound

$$E(b, u) = O(N^{1-1/(6r)}L^{-2/3}) = o(N/L^2). \quad (6.3)$$

By (4.8) and $b \log(4u) = F_N(u) - \log(4u)$,

$$\log A(b, u) \leq H_N(u) + E(b, u) + O(\log N). \quad (6.4)$$

Throughout I_N , $H_N''(u) \leq -L^2/(4rN)$. Also $u_* \in I_N$ and $u_* \geq N/(2L)$ eventually. If $u \in I_N$ but $|u/u_* - 1| \geq 1/L$, Taylor's theorem on the segment from u_* to u gives

$$M_r(N) - H_N(u) \geq \frac{1}{2} \frac{L^2}{4rN} \left(\frac{N}{2L^2} \right)^2 = \frac{N}{32rL^2}.$$

Combining with (6.3) and (6.4), eventually

$$\log A(b, u) \leq M_r(N) - \frac{N}{64rL^2}. \quad (6.5)$$

It remains to consider $|u/u_* - 1| < 1/L$. Write $u = u_*(1 + \varepsilon)$, so $|\varepsilon| < 1/L$. Since $t \asymp L$,

$$\alpha - \alpha_* = O(1), \quad \alpha_* := \frac{N/u_* - 1}{r} - \frac{1}{u_*} = \frac{\log(4u_*)}{r} - \frac{1}{2} - \frac{1}{u_*}.$$

Consequently $1 + \alpha_* \asymp L$ and $e^{-\alpha_*/3} \asymp u_*^{-1/(3r)}$. The bounded change in α changes its exponential factor only by a bounded multiplier. Thus

$$E(b, u) = O(u_*^{1-1/(3r)} L^{1/3}) = O(N^{1-1/(3r)} L^{-2/3+1/(3r)}) = O(R_r(N)). \quad (6.6)$$

Now (6.4) and $H_N(u) \leq M_r(N)$ prove the desired bound in this narrow interval. Sum it with (6.2) and (6.5) over at most N admissible values of u . Taking the logarithm costs at most $\log N$, which is absorbed in $R_r(N)$. This finishes Theorem 1.1.

7 Every fixed logarithmic order

We now convert the controlled main expression to polynomial expansions, including analytic error estimates. Retain L, ℓ, a, z from (1.6). The equation defining t is

$$t + \log t = L + a. \quad (7.1)$$

Define polynomials $p_j(z)$ successively, using a formal variable x , by

$$p_j(z) = -[x^j] \log \left(1 - zx + \sum_{k=1}^{j-1} p_k(z) x^{k+1} \right), \quad j \geq 1. \quad (7.2)$$

This is genuinely triangular: the coefficient of x^j involves only the earlier polynomials. Induction gives $\deg p_j \leq j$. Set

$$P_j(z) = p_j(z) + [x^{j-1}] \left(1 - zx + \sum_{k=1}^{j-1} p_k(z) x^{k+1} \right)^{-1}. \quad (7.3)$$

The first six coefficient polynomials are

$$\begin{aligned} P_1(z) &= z + 1, & P_2(z) &= z^2/2, \\ P_3(z) &= z^3/3 - z^2/2, & P_4(z) &= z^4/4 - 5z^3/6 + z^2/2, \\ P_5(z) &= z^5/5 - 13z^4/12 + 3z^3/2 - z^2/2, \\ P_6(z) &= z^6/6 - 77z^5/60 + 71z^4/24 - 7z^3/3 + z^2/2. \end{aligned} \quad (7.4)$$

The coefficient identities are reproducible by two independent exact-rational series algorithms in the package.

Lemma 7.1 (Analytic remainder for the reversion). *For each fixed $J \geq 1$,*

$$t = L - z + \sum_{j=1}^J p_j(z) L^{-j} + O((1 + |z|)^{J+1} L^{-J-1}). \quad (7.5)$$

Proof. Let T_J be the displayed finite expression without its error and set $U_J = T_J/L$. Then

$$T_J + \log T_J - L - a = \sum_{j=1}^J p_j(z) L^{-j} + \log U_J.$$

The recurrence (7.2) cancels all powers L^{-1} through L^{-J} . Put $D = 1 + |z|$. For fixed J , $D/L \rightarrow 0$, $T_J > 0$ eventually, and $U_J - 1 = O(D/L)$. The analytic Taylor remainder of the logarithm through degree J is $O((D/L)^{J+1})$. Every uncanceled monomial in its finite Taylor part has power of L^{-1} at least $J+1$, and degree in z at most that power, so its value has the same bound. Thus the full residual is $O(D^{J+1}L^{-J-1})$. The derivative of $t + \log t$ is $1 + 1/t \geq 1$ for $t > 0$. The mean value theorem transfers the residual bound to $T_J - t$. $\square$

Since $1/T_J = L^{-1}U_J^{-1}$, the reciprocal contribution at L^{-j} is the coefficient in (7.3). Substitution in M_r preserves the error in Lemma 7.1; the derivative of $t - 2 + 1/t$ is bounded near the large solution. The leading constant becomes $a - 2 + r/2 = \log 4 - 1$. By (1.5), the counting uncertainty is smaller than the fixed-order remainder. This proves (1.7).

For example,

$$t = L - z + \frac{z}{L} + \frac{z^2/2 - z}{L^2} + \frac{z^3/3 - 3z^2/2 + z}{L^3} + O\left(\frac{(1+|z|)^4}{L^4}\right),$$

$$\log A_s(n) = \frac{N}{r} \left[L - \ell + \log 4 - 1 + \frac{z+1}{L} + \frac{z^2}{2L^2} + \frac{z^3/3 - z^2/2}{L^3} \right] + O\left(\frac{N(1+\ell)^4}{L^4}\right).$$

Here $z+1 = \log \log N - \log 4 + r/2$. Replacing $N = n+1$ by n in any fixed smooth truncation changes its value by $O(\log n)$, because that truncation has derivative $O(\log n)$. This change is below the displayed remainders. We differentiate the explicitly given truncation, not an unknown counting error.

8 The controlled inverse to every fixed order

Differentiate the exact envelope $M_r(N) = H_N(u_*(N))$. The term involving du_*/dN vanishes because $H'_N(u_*) = 0$, so

$$M'_r(N) = \frac{\log(4u_*)}{r} = \frac{t-1+r/2}{r} \sim \frac{\log N}{r}. \quad (8.1)$$

This is positive eventually, and M_r tends to infinity. Hence $M_r(N_0) = Y$ has a unique sufficiently large solution.

Choose a sufficiently large fixed C and put $\Delta = CR_r(N_0)/\log N_0 = o(N_0)$. On shifts of this size, M'_r is comparable to $\log N_0$ and $R_r(N)$ is comparable to $R_r(N_0)$. The upper bound in Theorem 1.1 is strictly below Y at $N_0 - \Delta$ for large enough C , so monotonicity and integer rounding give

$$\nu_s(y) + 1 \geq N_0 - O(R_r(N_0)/\log N_0).$$

For the opposite side, the lower counting error is only $O(\log N)$. A sufficiently large fixed positive shift of N_0 raises M_r by more than this loss, by (8.1). Taking the next integer adds at most one. Thus

$$\nu_s(y) + 1 \leq N_0 + O(1).$$

This proves both (1.9) and (1.8). It uses explicit envelope derivatives and monotone bracketing, never derivatives of an unknown remainder.

The saddle equation gives the exact relation $N = e^{r/2}te^t/(4e)$. Substitute it in M_r to get

$$M_r(N) = K_r e^t \Pi_r(t), \quad K_r = \frac{e^{r/2}}{4er}, \quad \Pi_r(t) = t^2 + (r/2 - 2)t + 1.$$

This proves (1.10). The quadratics for $r = 1, 2$ are positive on the real line, and the derivative of $t + \log \Pi_r(t)$ is positive for large t , verifying the intended branch directly. Since $N_0 \sim rY/\log Y$, the counting uncertainty in (1.8) is equivalently

$$O\left(Y^{1-1/(3r)}(\log Y)^{-8/3+2/(3r)} + 1\right). \quad (8.2)$$

8.1 The first explicit inverse correction

Put $v = \log Y$, $k = \log v$, $A = r/2 - 2$, and $B = -\log K_r = 1 + \log(4r) - r/2$. The scalar equation becomes

$$t + 2 \log t + \log(1 + A/t + 1/t^2) = v + B. \quad (8.3)$$

With $s_0 = B - 2k$, expansion gives

$$t = v + s_0 + \frac{-2s_0 - A}{v} + O((1+k)^2/v^2).$$

The exact denominator in $N_0 = rY/(t + A + 1/t)$ is consequently

$$t + A + 1/t = v - 2k + (A + B) + \frac{4k - 2B - A + 1}{v} + O((1+k)^2/v^2).$$

The identities

$$A + B = \log(4r) - 1, \quad -2B - A + 1 = 1 + r/2 - 2 \log(4r)$$

produce the denominator in (1.11). Its omitted error is $O((1+k)^2/v^2)$, and both denominators are asymptotic to v , so the induced error in N_0 is $O(Y(1+k)^2/v^4)$. The much smaller counting error (8.2) and the shift -1 are absorbed at this explicit truncation scale. The shift remains present in the more precise controlled formula (1.8).

8.2 A recurrence for all inverse orders

Set $z = 2 \log v - B$ and use a formal variable x . When computing $q_j(z)$, put

$$U = 1 - zx + \sum_{k=1}^{j-1} q_k(z) x^{k+1}$$

and define

$$q_j(z) = -[x^j] \left\{ 2 \log U + \log(1 + Ax/U + x^2/U^2) \right\}. \quad (8.4)$$

This is triangular and $\deg q_j \leq j$. The first two are

$$q_1 = 2z - A, \quad q_2 = z^2 - (A + 4)z + A^2/2 + 2A - 1.$$

For every fixed J , the finite approximation

$$T_J = v - z + \sum_{j=1}^J q_j(z) v^{-j} \quad (8.5)$$

has residual $O((1 + \log v)^{J+1}/v^{J+1})$ in (8.3). To see this, divide T_J by v , use the recurrence to cancel the first J powers of v^{-1} , and expand the logarithms and reciprocals analytically near $(v^{-1}, T_J/v) = (0, 1)$. Here $z = O(\log v)$, and the same polynomial-degree argument as in Lemma 7.1 bounds the finite uncanceled terms. The derivative of the left side of (8.3) is

$$1 + \frac{2t + A}{t^2 + At + 1} \geq 1$$

for all sufficiently large positive t . The mean value theorem therefore proves

$$t_y = T_J + O((1 + \log v)^{J+1}/v^{J+1}).$$

Substitution into the denominator $t + A + 1/t$ gives the useful quantitative form

$$\nu_s(y) = \frac{rY}{T_J + A + 1/T_J} - 1 + O\left(\frac{Y(1 + \log v)^{J+1}}{v^{J+3}} + Y^{1-1/(3r)}v^{-8/3+2/(3r)} + 1\right). \quad (8.6)$$

Further expansion of the denominator or its reciprocal gives every prescribed fixed inverse-logarithmic order. The counting uncertainty lies below each such order. This completes Theorem 1.2.

9 Historical comparison and scope

The two size conventions are essential. Natural-size de Bruijn terms charge increasing size for deeper indices and have different asymptotics. Fixed index or unary-height bounds, linear or affine terms, typed terms, and beta-normal subclasses likewise do not settle these unrestricted sequences.

Bodini, Gardy, Gittenberger, and Jacquot [3] give explicit unrestricted bounds for the variable-size-one model, with exponential bases corresponding to $4n/(e \log n)$ and $9(1 + \varepsilon)n/(e \log n)$ raised to $n/2$, up to lower-order factors. These locate the first two logarithmic scales and leave a linear-size logarithmic gap. David, Grygiel, Kozik, Raffalli, Theyssier, and Zaionc [4] study the variable-size-zero convention and prove broad enumeration and structural estimates. Bodini, Gardy, Gittenberger, and Gołębiewski [5] analyze fixed abstraction counts and bounded unary-height structures. Their checked expanded 2015 preprint provides fixed-parameter singularity results, but no proved error uniform at the moving regime $u \asymp n/\log n$ used here.

[write] 5 October 2026. The literature statements of this section (the bases of Bodini–Gardy–Gittenberger–Jacquot’s Theorem 22, the results of David et al., and the absence of an error uniform at $u \asymp n/\log n$ in the 2015 preprint of Bodini–Gardy–Gittenberger–Gołębiewski) are reported as the manuscript states them. Neither the placement nor the write checked them against the papers; Part II, Section 18, makes further such statements, likewise unchecked.

The historical comparison is bounded to the located, checked sources. No claim of global priority is made. The present proof is an ordinary analytic proof accompanied by exact finite corroboration. It is not a Lean formalization and has not undergone external peer review. The report develops the all-logarithmic-order counting and inverse result; it does not repeat separate large-deviation results for binding geometry.

The approximation (1.4) permits a stretched-exponential multiplicative discrepancy between the full count and its envelope or the single-spine subclass. Thus it does not imply $A_s(n) \sim C(n)e^{M_r(n+1)}$ for any specified amplitude $C(n)$, a ratio limit between subclasses, or a full multiplicative transseries. Those conclusions need estimates below the logarithmic accuracy established here.

9.1 Concrete remaining questions

1. Determine the first scale and coefficient of the residual $\log A_s(n) - M_r(n+1)$ beyond all fixed inverse-logarithmic orders. The exponent $1 - 1/(3r)$ here is an upper-bound exponent from a shape-entropy optimization, not a proved asymptotic scale.
2. Determine the typical off-spine deficit $u-h$ near the moving saddle. The proof maximizes a majorant over deficits; that optimizing value is not a distributional law for actual random terms.
3. Obtain uniform coefficient asymptotics for the enriched spine class and its off-spine corrections, controlling the transition between fixed and growing abstraction count. Such control may identify an amplitude or a multiplicative equivalent.
4. Only after resolving those residual scales, ask whether an organized multiplicative expansion or transseries exists and which exponential sectors it requires.

These are research questions, not conclusions extracted by formal expansion of the current error term.

[write] **5 October 2026.** Status after the merge. Question 1 is open; its first sentence and the note after Theorem 1.1 explain why $R_r(N)$ is not a scale. Question 2 is Part II's question 2 (the typical size of $U_n - H_n$; for an actual term the largest unary depth h of a terminal leaf of its reduced tree is Part II's H_n); both are merged into item 2 of Section 20. Question 3 is merged with Part II's question 4 into item 3 of Section 9.2. Question 4 is item 4 there.

9.2 Further questions and research

[write] (5 October 2026.) Under Vladimir's standing rule of 4 October 2026, this subsection collects every claim of this Part that is stated without proof, with its source, the argument sketched for it and what is missing. Nothing in this Part was found to be wrong. The questions about random binding geometry, including the merged question on the deficit $u-h$, are in Section 20.

1. *A multiplicative equivalent, an amplitude, a multiplicative transseries* (Report 110: abstract, Section 9; Report 109: abstract, the remark at the end of Section 14, Section 18). Theorem 1.1 allows a stretched-exponential factor $\exp\{O(R_r(N))\}$ above $e^{M_r(N)}$ and a polynomial factor below it, so it implies no relation $A_s(n) \sim C(n)e^{M_r(n+1)}$, no ratio limit between subclasses and no amplitude. Sketch offered: uniform coefficient asymptotics of the enriched spine class (item 3) “may identify an amplitude or a multiplicative equivalent”. Missing: estimates below logarithmic accuracy.
2. *The first scale of $\log A_s(n) - M_r(N)$ beyond all fixed inverse-logarithmic orders, and its coefficient* (question 1). Known: the residual lies between $-O(\log N)$ and $O(R_r(N))$; the exponent $1 - 1/(3r)$ is that of a shape-entropy majorant (Proposition 4.2), not a proved scale, and nothing excludes a residual of order $\log N$. Evidence only, from Report 109's shipped exact tables (`data/109-binding-A135501_computed.txt` and the A220894 table), computed at the write: $(\log A_s(n) - M_r(N))/(N/\log N)$ is $-0.111, +0.014, +0.063, +0.089$ for $s = 1$ and $-0.387, -0.259, -0.201, -0.167$ for $s = 0$ at $n = 50, 100, 150, 200$; against Part II's Lambert expression the same quotient is $+0.342, +0.485, +0.542, +0.573$ ($s = 1$) and $+0.045, +0.192, +0.259, +0.299$ ($s = 0$), drifting towards the predicted $\frac{1}{2}$. These numbers prove nothing about the residual's order.
3. *Uniform coefficient asymptotics for the enriched spine class and its off-spine corrections*, controlling the transition between fixed and growing abstraction count (question 3), merged with Report 109's question 4: *can the weighted-placement sum* (12.3)

be controlled uniformly to successive orders, leading to a genuine expansion and a correspondingly sharper inverse? Partly answered: Theorem 1.1 gives every fixed inverse-logarithmic order of the logarithm of the total count, and Theorem 1.2 the corresponding inverse. Open: control of $A(b, u)$ itself, uniformly in (b, u) , beyond the leading exponential order of Remark 18.1, and anything at the multiplicative level.

4. *An organized multiplicative expansion or transseries, and its exponential sectors* (question 4), to be asked only after items 1–3.
5. *Explicit constants and onsets.* Both manuscripts state every O -term without constants or onset (Report 110 after Theorem 1.2 and in its delivered README; Report 109 in Section 19: the checks do not “certify an error constant, establish an eventual numerical threshold”). Missing: effective versions of Lemma 4.1, Proposition 4.2, the localization of Section 6 and the bracketing of Section 8; without them no finite threshold can be certified.
6. *The literature comparison* (Section 9; Part II, Section 18). Not checked at the placement or at the write: Bodini–Gardy–Gittenberger–Jacquot [3], Theorem 22 and Remark 23 (the bases $4n/(e \log n)$ and $9(1+\varepsilon)n/(e \log n)$); David et al. [4], Theorem 5.4 (abstractions between $n/(3 \log n)$ and $3n/\log n$ with high probability, and a unary-height lower bound); the 2015 preprint of [5] (no error uniform at $u \asymp n/\log n$); Gittenberger–Larcher [6] (the logarithm to second order in their introduction); Grygiel–Larcher [7] (the unrestricted problems called open); Bendkowski–Bodini–Dovgal [8] (the natural-size model). A standard web search at the placement found nothing beyond these sources. Both manuscripts disclaim global priority.

10 Reproducibility and verification

The analytic proof above does not depend on finite tests. The companion archive contains the TeX source, final PDF, standard-library exact checker, fixtures, deterministic build and integrity tools, corruption tests, and reader-facing validation summaries. It contains no downloaded source PDFs or third-party author code.

10.1 Three distinct kinds of evidence

1. **Analytic proof.** Sections 2–8 establish the asymptotic claims for all sufficiently large sizes, with all fixed orders justified by analytic remainders.
2. **Exact finite corroboration.** The standalone checker uses integers and rational numbers. It checks root and reduced-tree recurrences, size shifts and parity, reduced shape counts and height constraints, radical inequalities at positive rational parameters, spine convolution inequalities, probability moments, and forward and inverse series generated by independent algebraic routes. Its guards are explicit exceptions, so optimized Python does not disable them.
3. **Numerical diagnostics.** Numerical comparisons, if used during exploration, cannot prove eventual inequalities or uniform errors. None are used in the standalone exact checker or in the analytic proof; the packaged checks require neither floating-point arithmetic nor a symbolic-algebra package.

The standalone checker runs 17,643 exact finite checks. It generates forward coefficients through order eight and inverse coefficients through order six, and compares them with independently implemented formal identities. A separate implementation check and the

corruption campaign test that the validation guards reject altered logic and malformed or changed fixtures. The accompanying validation records state their finite scope; they do not replace the analytic arguments or constitute external peer review.

10.2 Local replay

Run the following commands from the extracted package directory:

```
python3 integrity.py
python3 -O integrity.py
python3 check.py
python3 -O check.py
python3 corruption_test.py
python3 build.py
python3 integrity.py
```

The mathematical checker has no non-standard Python dependencies. The PDF build requires pdfTeX and the standard LaTeX packages named in the source. The build fixes timestamps, suppresses changing PDF metadata, rebuilds from two clean directories, requires byte-identical results, and rejects overfull boxes or unresolved references. A fresh ZIP replay runs both Python modes, the corruption campaign, and a clean PDF rebuild, then verifies that the rebuilt PDF matches the packaged one and that the full file inventory is unchanged. The manifest checks payload integrity against recorded hashes, including the final PDF; it is not a mathematical certificate or a digital signature.

The corruption campaign changes mathematical logic, fixtures, manifest structure, and payload bytes in disposable copies. Mathematical mutants bypass the manifest intentionally, so a hash mismatch cannot hide an inactive mathematical guard. The normal and optimized interpreter runs must both reject them. The archive uses a fixed timestamp and sorted paths. Final checks include rendering and visual inspection of every PDF page; the QA images are not part of the mathematical evidence.

[write] 5 October 2026. In this repository the files named above are shipped under other names (`code/110-orders-check.py`, `data/110-orders-counts.json` and so on; the report's `README.md` maps them), and the scripts expect the delivered layout, so they run only on a copy of the delivered package. The delivered `report110.pdf`, `manifest.json` and `README.md` are not shipped; this `article.tex` replaces `report110.tex`, so the delivered build and replay scripts do not apply to it.

Part II

Logarithmic growth and random binding geometry

Bundle Report 109, archive `Closed_Lambda_Terms_Growth_and_Binding_Geometry_Source.zip`, delivered as `report109.tex` (16 pages), dated 2 October 2026. Delivered title “Unrestricted closed lambda terms: Logarithmic growth and random binding geometry”; author line “Report 109”. Printed as delivered, with `[write]` notes. Delivered Section k is Section $k + 10$ here, and its statement or equation $k.j$ is $(k + 10).j$; the proof of its Section 3 is printed in Part I (see the Guide). Remark 18.1 and Section 20 are added in the write.

Abstract of this Part (as delivered). We study two basic, unrestricted counting sequences for closed lambda terms up to alpha-equivalence: OEIS A135501, where every syntax vertex has size one, and A220894, where variable occurrences have size zero. A uniform positive-series majorant resolves four scales in their logarithms. If variables have size $s \in \{0, 1\}$ and $r = s + 1$, then

$$\log A_s(n) = \frac{n}{r} \left(\log n - \log \log n + \log 4 - 1 + \frac{\log \log n}{\log n} \right) + O\left(\frac{n}{\log n}\right).$$

The proof controls the critical values of a finite triangular family of generating functions, using a depth-forest inequality for reduced unary–binary trees. It does not require an unproved uniform extension of fixed-parameter singularity analysis. We also prove a controlled threshold inverse and a full joint large-deviation principle for the total number of abstractions and maximum unary depth. Both parameters are asymptotic to $n/\log n$ in every fixed finite L^p . The results sharpen the explicit historical estimates checked here, but do not establish a multiplicative equivalent, an all-orders transseries, or global priority. Exact, reproducible finite checks accompany the proofs and are kept logically separate from them.

Reading conventions. Where this Part’s letters differ from Part I’s:

This Part	Part I	Meaning
$w = W(4eN)$, $u_* = N/w$	$t = W(4eNe^{-r/2})$, $u_* = N/t$	different Lambert values and saddles
$t = u \log N/N$, T_n	—	scaled abstraction count (near 1)
$A_s(n, u)$	$A(b, u)$	the same count
$L = \log y$, $\ell = \log L$, $\log \ell$	Y, v, k	the inverse variables
$q = \log N$, $h = \log q$ (Section 14)	L, ℓ	logarithms of N
$G_N = F_N + u \log(2eN/u)$	$G_N = F_N + cu - \log 2$	different envelopes
H_n (random height)	h (reduced-tree height)	the same statistic
$L_m(z)$ (size series)	$T_m(x, z)$	z marks size here

11 The sequences and the main conclusions

11.1 Why this family is a useful asymptotic target

A lambda term is built from variables, abstraction $\lambda x.M$, and ordered application (MN) . Alpha-equivalence identifies consistent changes of bound-variable names. A term is closed when every variable occurrence is bound by an enclosing abstraction. These are fundamental objects in the combinatorics of binding syntax and random program generation. The two size conventions below give concrete, positive, eventually nondecreasing integer sequences with a clear exact recurrence; neither is an artificially oscillating or sporadic example.

Let a variable occurrence have size $s \in \{0, 1\}$, and let each abstraction and application have size one. Write $A_s(n)$ for the number of alpha-equivalence classes of closed terms of size n . Then

$$A_1(n) = \text{A135501}(n), \quad A_0(n) = \text{A220894}(n).$$

The index starts at $n = 0$. In particular, $\lambda x.x$ has size two for $s = 1$ and size one for $s = 0$. The beginning of the two sequences is

n	$A_1(n)$	$A_0(n)$
0	0	0
1	0	1
2	1	3
3	2	14
4	4	82
5	13	579
6	42	4741
7	139	43977
8	506	454283
9	1915	5159441
10	7558	63782411

All logarithms in this report are natural. Set

$$r = s + 1, \quad N = n + 1. \quad (11.1)$$

The shift N makes the exact vertex relation particularly simple.

11.2 Count and inverse

The positive real Lambert function W is defined by $W(z)e^{W(z)} = z$ for $z > 0$.

Theorem 11.1 (Four logarithmic scales). *For either $s = 0$ or $s = 1$, put $w = W(4eN)$. As $n \rightarrow \infty$,*

$$\log A_s(n) = \frac{N}{r} \left(w - 2 + \frac{1}{w} \right) + O\left(\frac{N}{\log N} \right). \quad (11.2)$$

Consequently,

$$\log A_s(n) = \frac{n}{r} \left(\log n - \log \log n + \log 4 - 1 + \frac{\log \log n}{\log n} \right) + O\left(\frac{n}{\log n} \right), \quad (11.3)$$

and

$$\lim_{n \rightarrow \infty} \frac{\log n}{n} A_s(n)^{r/n} = \frac{4}{e}. \quad (11.4)$$

Thus the coefficient of $n \log \log n / \log n$ is $1/r$. The coefficient of the next scale $n / \log n$ is not determined by this theorem. Expanding the Lambert expression to arbitrarily many orders would expand a lower-envelope model; it would not supply an all-orders expansion of the full count.

[write] **5 October 2026.** Superseded, and kept as a second route. Part I, Theorem 1.1, proves $M_r(N) - O(\log N) \leq \log A_s(n) \leq M_r(N) + O(R_r(N))$, with the shifted Lambert value $t = W(4eNe^{-r/2})$ and the extra term $N/2$ in $M_r(N)$; its error lies below every scale $N/(\log N)^k$. It implies (11.2)–(11.4): $M_r(N) - \frac{N}{r}(w - 2 + \frac{1}{w}) = \frac{N}{2 \log N} + O(N(\log \log N)^2/(\log N)^2)$, by the expansion

of Part I, Section 7, with $J = 1$ and that of Section 14 here. The coefficient of the next scale $n/\log n$, which the sentence above says this theorem does not determine, is determined by Part I, (1.7):

$$\log A_s(n) = \frac{n}{r} \left(\log n - \log \log n + \log 4 - 1 + \frac{\log \log n - \log 4 + r/2}{\log n} \right) + O\left(\frac{n(\log \log n)^2}{(\log n)^2}\right),$$

so the coefficient of $n/\log n$ is $\frac{1}{2} - \frac{\log 4}{r}$. The Lambert expression (11.2) has $-\frac{\log 4}{r}$ there: it is the lower envelope of the root-chain subclass, which misses the factor $e^{u/2+o(u)}$ of Remark 18.1. The last sentence above stays true: expanding (11.2) further expands that envelope, not the count; Part I expands $M_r(N)$ instead, whose error is below every fixed order.

Define the threshold inverse by

$$\nu_s(y) = \min\{n \in \mathbb{N} : A_s(n) \geq y\}. \quad (11.5)$$

Root abstraction is an injection from size n into size $n + 1$, so $A_s(n)$ is nondecreasing. Theorem 11.1 also implies that it tends to infinity.

Theorem 11.2 (Controlled inverse). *Let $L = \log y$, $\ell = \log L$, and $c_0 = \log 4 - 1$. Then*

$$\nu_s(y) = \frac{rL}{\ell - 2 \log \ell + \log r + c_0 + 4 \log \ell / \ell} + O\left(\frac{L}{\ell^3}\right). \quad (11.6)$$

The coefficient 4 is fixed at the displayed scale. A constant divided by ℓ in the denominator would change the answer at the unresolved error scale.

[write] **5 October 2026.** Superseded: Part I, Theorem 1.2, determines the constant divided by ℓ that the last sentence leaves open. In this Part's letters, (1.11) reads

$$\nu_s(y) = \frac{rL}{\ell - 2 \log \ell + \log r + c_0 + [4 \log \ell + 1 + r/2 - 2 \log(4r)]/\ell} + O\left(\frac{L(1 + \log \ell)^2}{\ell^4}\right),$$

since $\log r + c_0 = \log(4r) - 1$. The constant $1 + r/2 - 2 \log(4r)$ is about -1.273 for A220894 and -2.159 for A135501. It is not zero, so the right side of (11.6) differs from $\nu_s(y)$ by $r|1 + r/2 - 2 \log(4r)|L/\ell^3(1 + o(1))$: the error term of (11.6) is of exact order, and (11.6) lies below $\nu_s(y)$ for large y . Part I also gives the asymmetric bracket (1.9) and every fixed inverse order (Section 8).

Section 15 proves this by monotone bracketing; it never differentiates an unknown error term.

11.3 The geometry of a random term

Choose uniformly from the $A_s(n)$ closed alpha-classes. Let U_n be its number of abstractions, and let H_n be the maximum number of abstraction vertices on a root-to-leaf path. Binary vertices on the path do not count toward H_n . Put

$$T_n = \frac{U_n \log N}{N}, \quad X_n = \frac{H_n \log N}{N}. \quad (11.7)$$

Theorem 11.3 (Joint large deviations). *The pair (T_n, X_n) satisfies a large-deviation principle on $\mathbb{R}^2$, with speed n and good rate function*

$$J_r(t, x) = \begin{cases} (t - 1 - \log x)/r, & 0 < x \leq t, \\ +\infty, & \text{otherwise.} \end{cases} \quad (11.8)$$

Both marginals have rate

$$I_r(x) = \begin{cases} (x - 1 - \log x)/r, & x > 0, \\ +\infty, & x \leq 0. \end{cases} \quad (11.9)$$

Moreover, for every fixed $0 < p < \infty$,

$$T_n \longrightarrow 1 \quad \text{and} \quad X_n \longrightarrow 1 \quad \text{in } L^p.$$

In particular, $\mathbb{E}U_n \sim \mathbb{E}H_n \sim n/\log n$.

For clarity, a speed- n large-deviation principle with rate J means that every open set O and closed set F satisfy

$$\begin{aligned} \liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}((T_n, X_n) \in O) &\geq -\inf_O J, \\ \limsup_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}((T_n, X_n) \in F) &\leq -\inf_F J. \end{aligned}$$

“Good” means that all finite sublevel sets of J are compact.

Corollary 11.4 (Abstractions off a longest unary path). *The variable $D_n = (U_n - H_n) \log N/N$ has rate*

$$K_r(d) = d/r \quad (d \geq 0), \quad K_r(d) = +\infty \quad (d < 0).$$

For every $\delta > 0$,

$$\mathbb{P}(D_n \geq \delta) = \exp\{-n\delta/r + o(n)\}. \quad (11.10)$$

Hence a longest unary path contains all but $o(n/\log n)$ of the abstractions, in probability.

This concerns the entire root-to-leaf path. It does not say that the abstractions form a consecutive chain above the root.

[write] **5 October 2026.** Theorem 11.3 and Corollary 11.4 have no counterpart in Part I. Corollary 11.4 says that $U_n - H_n = o(n/\log n)$ in probability; its typical size is open (Section 20, item 2, which merges this Part’s question 2 with Part I’s). The comparison with David et al.’s Theorem 5.4 in Section 18 was not checked against that paper.

12 Exact combinatorics and the first sandwich

12.1 Binder positions rather than names

For a fixed unary–binary syntax tree, a leaf at unary depth d has exactly d available binders. Selecting one ancestor abstraction at each leaf determines exactly one closed alpha-class. Equivalently, this selects a valid de Bruijn index at each leaf. Distinct ancestor choices remain distinct under alpha-equivalence. Consequently the number of classes supported by a skeleton is the product of its leaf depths; a leaf of depth zero contributes zero.

If a term has u unary vertices, b binary vertices, and v leaves, then $v = b + 1$. Its size is

$$n = u + b + s(b + 1), \quad v = \frac{N - u}{r}, \quad b = \frac{N - u}{r} - 1. \quad (12.1)$$

Call u admissible at size n when $u \geq 1$ and $N - u$ is a positive multiple of r . For $r = 2$ this is a parity condition; it cannot be discarded in lower-bound constructions. Let $A_s(n, u)$ count classes with exactly u abstractions. The Catalan number

$$C_b = \frac{1}{b + 1} \binom{2b}{b}$$

counts plane full binary trees with b internal vertices.

Lemma 12.1 (Skeleton sandwich). *For every admissible pair (n, u) ,*

$$C_b u^{b+1} \leq A_s(n, u) \leq C_b \binom{u+2b}{u} u^{b+1}. \quad (12.2)$$

More exactly, if $\mathcal{B}_b$ denotes the plane full binary trees with b internal vertices, then

$$A_s(n, u) = \sum_{B \in \mathcal{B}_b} \sum_{\alpha_1 + \dots + \alpha_{2b+1} = u} \prod_{v \text{ leaf of } B} \left(\sum_{\alpha_i \geq 0} \alpha_i \right)_{i \text{ on the path to } v}. \quad (12.3)$$

Each path includes the root slot and the slot directly above its leaf.

Proof. Suppress unary vertices. Each remaining binary skeleton has $2b+1$ vertices and hence $2b+1$ slots directly above those vertices, including a root slot. Reinserting unary chains amounts to a weak composition of u into these slots, giving $\binom{u+2b}{u}$ placements. A leaf has at most u ancestors, proving the upper bound and the exact product formula. For the lower bound, place all u abstractions in the root slot. Every leaf then has u choices independently. There is no factorial for variable names. $\square$

We shall use the elementary estimates

$$\frac{4^b}{(2b+1)(b+1)} \leq C_b \leq 4^b. \quad (12.4)$$

The lower one follows because the middle binomial coefficient is the largest of the $2b+1$ coefficients whose sum is 4^b .

12.2 An exact recurrence

Let $L_m(z)$ count terms in a context with m distinguishable available binder positions. Variables, abstraction, and ordered application give the formal identity

$$L_m(z) = mz^s + zL_{m+1}(z) + zL_m(z)^2. \quad (12.5)$$

For $s=0$ its constant term is m ; this causes no infinite coefficient because every application or abstraction has positive size. With $a_s(n, m) = [z^n]L_m(z)$, the recurrence is

$$a_s(n, m) = m\mathbf{1}_{n=s} + \mathbf{1}_{n \geq 1} \left(a_s(n-1, m+1) + \sum_{i=0}^{n-1} a_s(i, m)a_s(n-1-i, m) \right). \quad (12.6)$$

In particular, $A_s(n) = a_s(n, 0)$. To compute through size M , only the triangular array $n+m \leq M$ is needed. All dependencies have smaller size. The code in the accompanying package also differentiates a unary marker to obtain exact first moments and independently checks them with the bivariate recurrence below.

[write] **5 October 2026.** (12.1) is Part I's (1.1). $L_m(z)$ here is the size series (z marks size); it is unrelated to Part I's single-spine series $L_u(x)$ (5.1). The lower half of the sandwich, the root-chain subclass, is the lower bound of this Part's counting theorem; Part I replaces it by the single-spine class.

13 A uniform critical value majorant

The crude placement factor in (12.2) is too large to resolve the fourth logarithmic scale. The key is to sum the zero-abstraction subtrees first and then bound the much smaller reduced-tree family.

Theorem 13.1 (Uniform coefficient bound). *Let $K = e(2 + \sqrt{3})$. For all admissible (n, u) ,*

$$C_b u^{b+1} \leq A_s(n, u) \leq 2uK^u(4u)^b. \quad (13.1)$$

The upper bound is uniform in both b and u .

[write] **5 October 2026.** The proof of this theorem is printed once, in Part I, Section 3: its upper half is Lemma 3.1, proved there by this Part's Sections 3.1–3.3 (Report 109's numbering), and its lower half is Lemma 12.1. This Part's equations (3.2)–(3.5) are Part I's (2.1), (2.2), (2.3) and the display $A(b, u) = [x^b]B_{0,u}(x)$ after them; its (3.6)–(3.11), Lemmas 3.2–3.3 and Remark 3.4 are Part I's (3.2)–(3.7), Lemmas 3.2–3.3 and the remark at the end of Section 3.3 there. Part I's Proposition 4.2 sharpens the upper half to $\log A(b, u) \leq b \log(4u) + u/2 + E(b, u) + O(\log(u + 1))$.

14 The moving saddle and four logarithmic scales

Define the real envelope

$$F_N(u) = \frac{N - u}{r} \log(4u), \quad 0 < u < N. \quad (14.1)$$

It is strictly concave, since

$$F'_N(u) = \frac{N/u - 1 - \log(4u)}{r}, \quad F''_N(u) = -\frac{N/u^2 + 1/u}{r} < 0. \quad (14.2)$$

Its maximizer is $u_* = N/w$, with $w = W(4eN)$, and direct substitution gives

$$F_N(u_*) = \frac{N}{r} \left(w - 2 + \frac{1}{w} \right). \quad (14.3)$$

For sufficiently large n , a nearest admissible integer lies within $r/2$ of u_* . The saddle is far from either endpoint, so this choice has $b \geq 0$. Taylor's theorem and (14.2) give a lattice loss $O((\log N)^2/N) = o(1)$. The lower Catalan estimate in (12.4) then yields

$$\log A_s(n) \geq F_N(u_*) - O(\log N). \quad (14.4)$$

Let $c = \log K$. The uniform upper bound simplifies exactly to

$$\log A_s(n, u) \leq F_N(u) + cu - \log 2. \quad (14.5)$$

There is no additional $\log u$ term: it has already been absorbed into F_N using $b+1 = (N-u)/r$. The maximizer u_c of $F_N(u) + cu$ is, for large N , interior and satisfies

$$u_c = N/x_c, \quad x_c + \log x_c = \log(4eN) - rc, \quad x_c = W(4eNe^{-rc}) \sim \log N.$$

Thus $u_c = O(N/\log N)$, and

$$F_N(u_c) + cu_c \leq F_N(u_*) + O(N/\log N).$$

Summing at most N admissible values costs only $O(\log N)$. Together with (14.4), this proves (11.2).

[write] **5 October 2026.** This proof is the second, weaker route to the count. Its lower bound uses the root-chain subclass at Part II's saddle $u_* = N/w$ ((14.4)), and its upper bound gives away $cu = O(N/\log N)$; Part I gains the factor $e^{u/2}$ in the lower bound by the single-spine class and controls the upper bound to $O(R_r(N))$ by the height-sensitive majorant and two localization stages.

Write $q = \log N$ and $h = \log q$. The identity $w + \log w = q + 1 + \log 4$ gives, by substitution,

$$w = q - h + 1 + \log 4 + \frac{h - 1 - \log 4}{q} + O(h^2/q^2),$$

$$w - 2 + w^{-1} = q - h + \log 4 - 1 + \frac{h - \log 4}{q} + O(h^2/q^2).$$

Since $h^2/q \rightarrow 0$, the last remainder is $o(1/q)$. The constant $-\log 4/q$ lies within the unresolved $O(1/q)$ error for the full count, but h/q does not. Replacing N by n costs $O(\log n)$ and proves (11.3). Dividing that equation by n/r and exponentiating proves (11.4).

[write] **5 October 2026.** For the full count the constant at this scale is determined by Part I: inside the bracket the coefficient of $1/q$ is $h - \log 4 + r/2$, not $h - \log 4$ ((1.7) with $J = 1$; Part I's $z + 1 = \ell - \log 4 + r/2$). The extra $r/2$ is the enhancement $u/2$ at $u \approx N/\log N$ (Remark 18.1).

Remark 14.1 (What exponential precision is available). A correct alternative reading of Theorem 11.1 is

$$A_s(n) = \exp\left\{\frac{n}{r} \left(\log n - \log \log n + \log 4 - 1 + \frac{\log \log n}{\log n} \right) + O\left(\frac{n}{\log n}\right)\right\}.$$

The residual factor can be as large as $\exp\{O(n/\log n)\}$. It cannot be replaced by $1 + o(1)$, a constant amplitude, or a power of n on the basis of this proof.

[write] **5 October 2026.** Relative to the Lambert form (11.2) the residual factor is in fact of exact order $\exp\{(\frac{1}{2} + o(1))N/\log N\}$: by Part I, $\log A_s(n) - \frac{N}{r}(w - 2 + \frac{1}{w}) = \frac{N}{2\log N}(1 + o(1))$. Relative to the four-scale form displayed in this remark it is $\exp\{(\frac{1}{2} - \frac{\log 4}{r} + o(1))n/\log n\}$, also of exact order because $\frac{1}{2} - \frac{\log 4}{r} \neq 0$ (about -0.193 for $r = 2$ and -0.886 for $r = 1$; the note after Theorem 11.1). Evidence only, from the exact counts of the independent check: the four-scale residual divided by $n/\log n$ is -0.901 for $s = 0$ at $n = 600$, and -0.019 for $s = 1$ (limit -0.193), where the convergence is still slow at that size. Relative to Part I's $e^{M_r(N)}$ it lies between $\exp\{-O(\log N)\}$ and $\exp\{O(R_r(N))\}$; this still gives no factor $1 + o(1)$, constant amplitude or power of n (Part I, Section 9). (Dated note, 5 October 2026: the first sentence of this note first read "Relative to the expression of Theorem 11.1, the residual factor is in fact of exact order $\exp\{(\frac{1}{2} + o(1))N/\log N\}$ ". That is true of the Lambert form (11.2), but the remark displays the four-scale form (11.3), against which the constant is $\frac{1}{2} - \frac{\log 4}{r}$; corrected after the independent check recorded at the end of Section 20.)

15 The threshold inverse

Set

$$f(x) = \frac{x}{r} \left(\log x - \log \log x + c_0 + \frac{\log \log x}{\log x} \right).$$

Theorem 11.1 says that, at integer arguments,

$$\log A_s(n) = f(n) + O(n/\log n). \quad (15.1)$$

For $L = \log y$, $\ell = \log L$, let $k = \log \ell$, $a = \log r + c_0$, and

$$D = \ell - 2k + a + 4k/\ell, \quad n_0 = rL/D.$$

As $y \rightarrow \infty$, $D \sim \ell$ and $n_0 \rightarrow \infty$. Expanding only the explicit smooth function, write $v = \log n_0$ and obtain

$$\begin{aligned} v &= \ell - k + \log r + (2k - a)/\ell + O(k^2/\ell^2), \\ \log v &= k + (-k + \log r)/\ell + O(k^2/\ell^2), \\ \frac{\log v}{v} &= k/\ell + O(k^2/\ell^2). \end{aligned}$$

Therefore

$$v - \log v + c_0 + \frac{\log v}{v} = D - \frac{a + \log r}{\ell} + O(k^2/\ell^2) = D + O(1/\ell),$$

where $k^2/\ell \rightarrow 0$. It follows that

$$f(n_0) - L = O(L/\ell^2). \quad (15.2)$$

The count error in (15.1) is also $O(L/\ell^2)$ locally, because $n_0 \asymp L/\ell$ and $\log n_0 \sim \ell$.

For completeness, make the bracketing step explicit. Choose a fixed sufficiently large $M > 0$ and integers

$$n_- = \lfloor n_0 - ML/\ell^3 \rfloor, \quad n_+ = \lceil n_0 + ML/\ell^3 \rceil.$$

The derivative of the explicit f satisfies $f'(x) \sim \ell/r$ uniformly between these points. The mean value theorem therefore makes $f(n_\pm) - f(n_0)$ have opposite signs and magnitudes asymptotic to $ML/(r\ell^2)$. Choose M to dominate the fixed constants in (15.1) and (15.2). Then for all sufficiently large y ,

$$\log A_s(n_-) < L \leq \log A_s(n_+).$$

Monotonicity gives $n_- < \nu_s(y) \leq n_+$. Rounding contributes $O(1)$, negligible relative to L/ℓ^3 . This proves Theorem 11.2 without any regularity assumption on the unknown error.

The denominator form cleanly separates the known scales. In particular, the simpler expansion

$$\nu_s(y) = \frac{rL}{\ell} \left[1 + \frac{2 \log \ell - \log r - c_0}{\ell} + O\left(\frac{(\log \ell)^2}{\ell^2}\right) \right]$$

is valid but hides the improvement supplied by the determined $4 \log \ell/\ell$ denominator term.

[write] **5 October 2026.** Superseded by Part I, Theorem 1.2 (see the note after Theorem 11.2); the bracketing method, monotonicity with an explicit smooth function and no derivative of an unknown error, is the same as Part I's. Letters here: $L = \log y$, $\ell = \log L$, $k = \log \ell$, D the denominator, $n_0 = rL/D$, $v = \log n_0$; Part I's are Y , v , $k = \log v$ and N_0 .

16 Total abstraction deviations

We first establish the marginal estimates needed to control noncompact regions in the joint problem. These use only the weaker $o(n)$ part of Theorem 11.1.

From the skeleton sandwich and $u + 2b \leq 2N$, the inequality $\binom{M}{u} \leq (eM/u)^u$ gives

$$\log A_s(n, u) \leq G_N(u) - \log 4, \quad G_N(u) = F_N(u) + u \log(2eN/u). \quad (16.1)$$

For $u \log N/N$ in a fixed compact subset of $(0, \infty)$, the last term is $O(N \log \log N / \log N) = o(N)$. The lower bound differs from $F_N(u)$ by only $O(\log N)$. Hence, uniformly in such a compact interval,

$$\frac{1}{n} \log \mathbb{P}(U_n = u) = -I_r\left(\frac{u \log N}{N}\right) + o(1). \quad (16.2)$$

Indeed, putting $t = u \log N/N$,

$$F_N(u) = \frac{N}{r} (\log N - \log \log N + \log 4 + \log t - t) + o(N).$$

Subtract the total-count logarithm to get (16.2).

The envelope G_N is strictly concave. If its maximizer is N/x , its stationary equation is

$$x + (r + 1) \log x = \log N + 1 + (2 - r) \log 2. \quad (16.3)$$

This has a unique positive solution, with $x \sim \log N$, so the scaled maximizing unary count tends to one. For every fixed $0 < a < 1 < b$, the envelope is eventually increasing up to $aN/\log N$ and decreasing beyond $bN/\log N$. Bound each tail by at most N copies of the appropriate endpoint envelope. A single admissible integer within distance r on the correct side supplies the matching lower bound. Thus

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}(T_n \leq a) = -I_r(a), \quad (16.4)$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}(T_n \geq b) = -I_r(b). \quad (16.5)$$

In particular, $I_r(a) \rightarrow \infty$ as $a \downarrow 0$ and $I_r(b) \rightarrow \infty$ as $b \rightarrow \infty$. These provide the lower-boundary exclusion and exponential tightness. The local estimate gives the open lower bound and compact upper bound, while the two tails extend the latter to every closed set. This proves the marginal LDP for T_n .

[write] **5 October 2026.** The envelope G_N of (16.1) is not Part I's coarse envelope G_N of (6.1), although both are F_N plus a term of order $N/\log N$ at the saddle.

17 The joint abstraction and height principle

17.1 A local construction that reaches the boundary

Let $A_s(n, u, h)$ count terms with exactly u abstractions and maximum unary height h . Every leaf now has at most h binders, so

$$A_s(n, u, h) \leq C_b \binom{u + 2b}{u} h^{b+1}. \quad (17.1)$$

For a lower bound, suppose

$$h \geq 2, \quad 1 \leq u - h + 1 \leq b + 1. \quad (17.2)$$

Start with any plane full binary tree with b internal vertices. Put $h - 1$ unary vertices in its root slot, then put one unary vertex directly above each of the first $u - h + 1$ leaves in left-to-right order. This gives exactly u unary vertices and height exactly h . Every leaf has at least $h - 1$ binders, hence

$$A_s(n, u, h) \geq C_b (h - 1)^{b+1}. \quad (17.3)$$

Distinct binary trees and binder selections remain distinct terms.

Consider any compact subset of $0 < x \leq t < \infty$ bounded away from $x = 0$, with

$$t = u \log N/N, \quad x = h \log N/N.$$

Conditions (17.2) hold uniformly for all sufficiently large n : $u, h = O(N/\log N)$, $h \rightarrow \infty$, and $b+1 \sim N/r$. The construction also works on $x = t$ by taking $h = u$, so it loses neither the cone boundary nor the $r = 2$ parity subsequence.

The loss from replacing h with $h - 1$ is

$$(b+1) \log \frac{h}{h-1} = O(\log N).$$

The Catalan correction is $O(\log N)$ and the placement entropy in (17.1) is $o(N)$. Consequently,

$$\log A_s(n, u, h) = \frac{N}{r} (\log N - \log \log N + \log 4 + \log x - t) + o(N), \quad (17.4)$$

$$\frac{1}{n} \log \mathbb{P}(U_n = u, H_n = h) = -J_r(t, x) + o(1), \quad (17.5)$$

uniformly on the indicated compacts.

17.2 The singular small-height boundary

Local estimates alone do not prove the full two-dimensional LDP because x can approach zero. Fix $0 < a < 1$ and $h = \lfloor aN/\log N \rfloor$. For the event $H_n \leq h$, each leaf has at most $\min(u, h)$ choices. For $u \leq h$, the logarithmic upper envelope is $G_N(u) - \log 4$ from (16.1), which is maximized at h for large N .

For $u \geq h$, use instead

$$Q_h(u) = \frac{N-u}{r} \log(4h) + u \log(2eN/u).$$

Its derivative is

$$Q'_h(u) = -\frac{1}{r} \log(4h) + \log(2eN/u).$$

At $u = h$, the first term is $-\log N/r + O(\log \log N)$ whereas the second is only $O(\log \log N)$. The derivative is therefore negative for large N and decreases further with u . The two envelopes agree at h , so summing at most N values of u gives

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}(X_n \leq a) \leq -I_r(a).$$

For the matching lower bound, take the largest admissible $u \leq h$. Then $h - u < r$. Count the entire root-chain subclass, with $C_b u^{b+1}$ terms, all of height $u \leq h$. The scaled u tends to a , so the same logarithmic expansion gives

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}(X_n \leq a) = -I_r(a). \quad (17.6)$$

As $a \downarrow 0$, these rates tend to infinity.

17.3 Globalization and contractions

Because $0 < X_n \leq T_n$, the upper tail (16.5) supplies exponential tightness of the pair. Every point of $0 < x \leq t$ admits admissible lattice approximations; on $x = t$ choose $h = u$. Equation (17.5) gives the open-set lower bound. On a compact set separated from $x = 0$, the same uniform estimate and at most N^2 lattice pairs give the compact-set upper bound.

For a general closed set, split the event into a compact middle region, $\{X_n \leq a\}$, and $\{T_n \geq b\}$. Apply the compact bound, (17.6), and (16.5), then let $a \downarrow 0$ and $b \rightarrow \infty$. This

proves the closed-set bound. The rate is lower semicontinuous, including at the cone and at $x = 0$. Moreover,

$$J_r(t, x) = I_r(x) + (t - x)/r \geq I_r(x).$$

A finite sublevel set keeps x away from zero and infinity; the identity $t = rJ_r(t, x) + 1 + \log x$ then bounds t . Hence the rate is good, proving the joint LDP.

For fixed x , minimizing $J_r(t, x)$ over $t \geq x$ gives $t = x$ and rate $I_r(x)$. For fixed t , minimizing over $0 < x \leq t$ likewise gives $x = t$. For the difference $d = t - x \geq 0$,

$$J_r(x + d, x) = \frac{d + x - 1 - \log x}{r},$$

whose minimum over $x > 0$ occurs at $x = 1$ and equals d/r . The contraction principle proves both marginal rates and the deficit rate. Since K_r is continuous for $d > 0$, the closed tail $[\delta, \infty)$ and open tail (δ, ∞) have equal infimum δ/r , proving (11.10).

Both marginal rates have unique zero at one. For any fixed $\varepsilon > 0$, each fixed relative deviation has exponentially small probability. Since $T_n, X_n \leq \log N$, its contribution to the p th absolute centered moment is at most $(\log N + 1)^p$ times that exponentially small probability. Inside the ε window it is at most ε^p . Let $n \rightarrow \infty$ and then $\varepsilon \downarrow 0$ to prove the stated L^p convergence.

Finally, changing the normalization from $\log(n + 1)/(n + 1)$ to $\log n/n$ changes either normalized variable by at most $O(\log n/n)$ deterministically, since $U_n, H_n \leq n$. This is exponential equivalence at every speed, so every stated LDP and limit also holds with the simpler normalization.

18 Historical comparison and the unresolved frontier

18.1 What the checked sources establish

Bodini, Gardy, Gittenberger, and Jacquot [3] treat the unrestricted size-one class in Section 5. Their Theorem 22 bounds its growth between expressions with principal bases $4n/(e \log n)$ and $9(1 + \varepsilon)n/(e \log n)$, raised to the power $n/2$, with further lower-order factors. These identify the first two logarithmic scales but leave a linear-in- n logarithmic gap. Theorem 11.1 fixes that linear term and one additional logarithmic scale.

For variable size zero, David, Grygiel, Kozik, Raffalli, Theyssier, and Zaionc [4] establish broad enumeration bounds and structural results. In particular, their Theorem 5.4 places the abstraction count, with high probability, between $n/(3 \log n)$ and $3n/\log n$, and gives a unary-height lower bound. Theorem 11.3 supplies the limiting constant one for both parameters and the complete rate functions in the present unrestricted model. This comparison does not purport to subsume their normalization or head-abstraction theorems.

The expanded preprint of Bodini, Gardy, Gittenberger, and Gołębiewski [5] proves fixed-abstraction singularity asymptotics and studies bounded unary height and bounded indices. It also describes empirical unary-height growth of order $n/\log n$. A fixed- u asymptotic cannot be inserted into a saddle with $u \sim n/\log n$ without a uniform error estimate. The proof in Section 13 provides its own finite, uniform majorant instead.

The introduction of Gittenberger and Larcher [6] describes the unrestricted logarithm through its second-order term. Grygiel and Larcher [7] explicitly describe the unrestricted size-zero and size-one enumeration problems as open; their own results impose a fixed index bound. These directly relevant sources were checked, together with targeted searches for later refinements through the report date. This is a bounded comparison, not a certificate that no earlier, unpublished, or poorly indexed matching result exists.

The natural-size model studied by Bendkowski, Bodini, and Dovgal [8] charges for the size of a de Bruijn index. It has different enumeration and probability laws. Likewise, linear, affine, beta-normal, simply typed, bounded-index, and bounded-height classes are not substitutes for A135501 or A220894.

[write] **5 October 2026.** The statements of this subsection about the cited papers (Theorem 22 and Remark 23 of [3], Theorem 5.4 of [4], the 2015 preprint of [5], the introduction of [6], the “open” of [7], and [8]) are reported as delivered; they were not checked against the papers at the placement or at the write (Section 9.2, item 6).

18.2 Why the result is not a complete transseries

The two decisive sources of information are an exact lower subclass and a uniform upper estimate whose logarithmic ratio is $O(u + \log(b+1))$. At the saddle, $u \asymp n/\log n$, so this leaves precisely the error scale in Theorem 11.1. Nothing here determines the actual weighted-placement enhancement at that scale.

[write] **5 October 2026.** Determined at the moving saddle by Part I. Part I’s Proposition 4.2 replaces the logarithmic ratio $O(u + \log(b+1))$ by $u/2 + E(b, u) + O(\log(b+u+1))$ with $E(b, u) = o(u)$ at the saddle, and its single-spine bound (5.6) gives $u/2 - O(\log u)$ from below; the weighted-placement enhancement over the root-chain subclass is therefore $e^{u/2+o(u)}$ there (Remark 18.1). It accounts for the missing coefficient of $n/\log n$ (note after Theorem 11.1).

A concrete next object is

$$R(b, u) = \frac{A_s(n, u)}{C_b u^{b+1}}, \quad n = u + rb + s. \quad (18.1)$$

The ratio is independent of the size convention once (b, u) is fixed. The present proof supplies an upper bound exponential in u , up to a polynomial factor in b , but no asymptotic formula for $\log R(b, u)/u$ in the moving regime. Establishing its limit, with adequate uniformity, would be a natural route to the missing $n/\log n$ coefficient. A full equivalent would require far sharper control, including effective prefactors and uniform summation errors.

Remark 18.1 ([write] The ratio frontier to leading order; 5 October 2026). Neither manuscript states this; it combines two estimates of Part I. Let $R(b, u)$ be as in (18.1), write $A(b, u) = A_s(n, u)$, $\alpha = b/u$, $H_u = \sum_{j=1}^u 1/j$, and let $E(b, u) = 3 \cdot 2^{1/3} u (1 + \alpha)^{1/3} e^{-\alpha/3}$ be as in (4.6).

1. For all integers $u \geq 1$ and $b \geq 0$,

$$0 \leq \log R(b, u) \leq \frac{u}{2} + E(b, u) + \log(2u(u+1)(2b+1)(b+1)). \quad (18.2)$$

2. For every $a > 0$ and all integers $u \geq 1$, $b \geq 0$ with $b \geq \frac{u}{2}(H_u - 1) + au$,

$$\log R(b, u) \geq \frac{u}{2} - \frac{1}{4} \log u - \frac{1}{2} - \log(1 + a^{-2}). \quad (18.3)$$

3. Consequently $\log R(b, u)/u \rightarrow \frac{1}{2}$ along every sequence of pairs with $u \rightarrow \infty$, $b \geq \frac{u}{2}(H_u - 1) + au$ for one fixed $a > 0$, and $\log(b+1) = o(u)$.
4. *At the moving saddle.* Let t and $u_* = N/t$ be Part I’s (1.2), and let $u = u_N$ be admissible with $(u - u_*) \log N/u_* \rightarrow 0$ (for example the admissible u_0 within $r/2$ of u_* chosen in Section 5), with $b = (N - u)/r - 1$. Then

$$-O(\log u) \leq \log R(b, u) - \frac{u}{2} \leq O(u^{1-1/(3r)} (\log u)^{1/3}), \quad (18.4)$$

so that $\log R(b, u)/u \rightarrow \frac{1}{2}$.

Proof. (1) The lower bound $R \geq 1$ is the left inequality of (12.2). For the upper bound, Proposition 4.2, (4.7), gives $A(b, u) \leq 2u^2(u+1)(4u)^b \exp\{u/2 + E(b, u)\}$ for all $u \geq 1$, $b \geq 0$, and (12.4) gives $C_b \geq 4^b/((2b+1)(b+1))$. Dividing, $R(b, u) \leq 2u(u+1)(2b+1)(b+1) \exp\{u/2 + E(b, u)\}$.

(2) Part I's inequality (5.6) gives $R(b, u) \geq Q_u(\rho) \mathbb{P}(J \leq b)$, with $Q_u(\rho) = \sqrt{u^u/u!}$ by (5.2). Since $\log$ is concave, $\int_{j-1}^j \log x \, dx \geq \frac{1}{2}(\log(j-1) + \log j)$ for $j \geq 2$; summing over $2 \leq j \leq u$ gives $u \log u - u + 1 \geq \log u! - \frac{1}{2} \log u$, also for $u = 1$. Hence $\log Q_u(\rho) = \frac{1}{2}(u \log u - \log u!) \geq \frac{u}{2} - \frac{1}{2} - \frac{1}{4} \log u$. By (5.4) and (5.5), J has mean $\mu = \frac{u}{2}(H_u - 1)$ and variance $\sigma^2 \leq u^2$. If $b \geq \mu + au$, then $J > b$ implies $J - \mu > au$, so Cantelli's inequality, proved in Section 5 in the form $\mathbb{P}(X \geq c) \leq \sigma^2/(\sigma^2 + c^2)$ for $c > 0$, gives

$$\mathbb{P}(J > b) \leq \frac{\sigma^2}{\sigma^2 + a^2 u^2} \leq \frac{u^2}{u^2 + a^2 u^2} = \frac{1}{1 + a^2},$$

because $\sigma^2 \mapsto \sigma^2/(\sigma^2 + c^2)$ is increasing. Thus $\mathbb{P}(J \leq b) \geq a^2/(1 + a^2) = 1/(1 + a^{-2})$, and (18.3) follows.

(3) Since $H_u \geq \log(u+1) \rightarrow \infty$, the hypothesis gives $\alpha \geq \frac{1}{2}(H_u - 1) + a \rightarrow \infty$, hence $E(b, u)/u = 3 \cdot 2^{1/3} (1 + \alpha)^{1/3} e^{-\alpha/3} \rightarrow 0$. The logarithmic terms in (18.2) and (18.3) are $O(\log(u+1) + \log(b+1)) = o(u)$. Dividing both inequalities by u gives the limit $\frac{1}{2}$.

(4) Put $L = \log N$. By (5.8), $N/u_* = t = \log(4u_*) + 1 - r/2$ and $t \sim L$, so $u_* \sim N/L$, and $u \sim u_*$ by hypothesis. Write $\alpha_* = (N/u_* - 1)/r - 1/u_* = \log(4u_*)/r - \frac{1}{2} - 1/u_*$, as in Section 6. Then $\alpha = (N/u - 1)/r - 1/u$ and

$$\left| \frac{N}{u} - \frac{N}{u_*} \right| = t \frac{|u - u_*|}{u} = O\left(L \cdot \frac{|u - u_*|}{u_*}\right) \rightarrow 0,$$

so $\alpha = \alpha_* + o(1)$. With $H_u \leq 1 + \log u$ (Section 5) and $\log u = \log u_* + o(1)$,

$$\alpha - \frac{1}{2}(H_u - 1) \geq \left(\frac{1}{r} - \frac{1}{2}\right) \log u_* + \frac{\log 4}{r} - \frac{1}{2} + o(1).$$

For $r = 2$ the right side tends to $\log 2 - \frac{1}{2} > 0$, and for $r = 1$ to $+\infty$; so for large N the hypothesis of (2) holds with $a = a_0 = (\log 2 - \frac{1}{2})/2$, the constant of Section 5, and (18.3) gives the left inequality of (18.4). For the right one, $\alpha = \log(4u_*)/r - \frac{1}{2} + o(1)$ gives $e^{-\alpha/3} = O(u^{-1/(3r)})$ and $(1 + \alpha)^{1/3} = O((\log u)^{1/3})$, so $E(b, u) = O(u^{1-1/(3r)}(\log u)^{1/3})$; and $b < N$ with $\log N = O(\log u)$ makes the last term of (18.2) $O(\log u)$. Finally $|u_0 - u_*| \leq r/2$ gives $(u_0 - u_*)L/u_* = O(L^2/N) \rightarrow 0$, so u_0 is an example. $\square$

[write] 5 October 2026. What the remark settles and what it does not. The paragraph above asks for the limit of $\log R(b, u)/u$ in the moving regime and calls it a natural route to the missing coefficient of $n/\log n$. Part (4) gives that limit, $\frac{1}{2}$, with the error $O(u^{1-1/(3r)}(\log u)^{1/3})$, which at the saddle is $O(R_r(N))$, the error of Part I's Theorem 1.1. Part I's proof of that theorem is the completed route: $\log(C_b u^{b+1}) + u/2 = H_N(u) - O(\log N)$ is its envelope (5.7). The remark does not by itself reprove Theorem 1.1: the global upper bound needs Part I's localization (Section 6) for u away from the saddle, where the hypothesis of (3) can fail. Not settled: the second-order behaviour of $\log R(b, u) - \frac{u}{2}$ at the saddle (between $-O(\log u)$ and $O(u^{1-1/(3r)}(\log u)^{1/3})$), and the limit for bounded α , or for $\alpha \rightarrow \infty$ with $\alpha - \frac{1}{2}(H_u - 1)$ not bounded below by a positive constant (for $r = 2$ this includes part of Part I's window $|u/u_* - 1| < 1/\log N$, where $|\alpha - \alpha_*|$ can approach $\frac{1}{2}$); see Section 20, item 5. Evidence only, from the exact recurrence (2.3) computed at the write for $r = 2$ at the admissible u nearest u_* : $\log R(b, u)/u = 0.737, 0.738, 0.732$ at $(b, u) = (13, 10), (25, 16), (38, 22)$ ($N = 38, 68, 100$), and the single-spine part $[x^b]L_u/(C_b u^{b+1})$ alone gives 0.576, 0.568, 0.559; both lie above $\frac{1}{2}$; the single-spine values decrease slowly, while the full values are not yet monotone at these sizes (the independent check, computing to $u = 120$, found $\log R/u$ rising to 0.754 at $N = 72$ and then decreasing to 0.664 at $N = 740$), as the error term allows. These values neither confirm nor refute the rate. (Dated note,

5 October 2026: the sentence before last first read “both lie above $\frac{1}{2}$ and decrease slowly”, which the full values 0.737, 0.738, 0.732 do not; corrected after the independent check recorded at the end of Section 20. Since that check, Proposition 18.2 below settles the limit $\frac{1}{2}$ whenever $\alpha - \frac{1}{2} \log \log u \rightarrow \infty$ and $\log(b+1) = o(u)$, in particular whenever $\alpha - \frac{1}{2}(H_u - 1)$ is bounded below, and gives the bound (18.4) in the whole window for $r = 2$; what remains open is re-scoped in Section 20, item 5.)

Proposition 18.2 ([write], independent check; 5 October 2026). *Let $R(b, u)$, $\alpha = b/u$ and H_u be as in Remark 18.1. For real $K \geq 0$ let κ_K be the least positive integer with $H_{\kappa_K} \geq 2K + 3$; then $\kappa_K \leq e^{2K+3}$ (for example $\kappa_0 = 11$ and $\kappa_1 = 83$). For all real $K \geq 0$ and all integers $u \geq 1$ and $b \geq 0$ with $b \geq \frac{u}{2}(H_u - 1) - Ku$,*

$$\log R(b, u) \geq \frac{u}{2} - 2\kappa_K(1 + \log u). \quad (18.5)$$

Consequently:

1. $\log R(b, u)/u \rightarrow \frac{1}{2}$ along every sequence of pairs with $u \rightarrow \infty$, $\log(b+1) = o(u)$ and $\alpha - \frac{1}{2} \log \log u \rightarrow \infty$. This includes $\alpha \geq \delta \log u$ for one fixed $\delta > 0$, and $\alpha \geq \frac{1}{2}(H_u - 1) - K$ for one fixed $K \geq 0$, so it extends Remark 18.1(3);
2. for $r = 1$ and $r = 2$ the two-sided bound (18.4) holds for all admissible u in Part I's window $|u/u_* - 1| < 1/\log N$ (Section 6), with $b = (N - u)/r - 1$ and constants depending only on r ; this extends Remark 18.1(4), whose hypothesis $(u - u_*) \log N/u_* \rightarrow 0$ excludes, for $r = 2$, part of that window.

Proof. Write $q_j = [x^j]Q_u(x) \geq 0$, so that $\mathbb{P}(J = j) = q_j \rho^j / Q_u(\rho)$ by (5.3). The convolution argument for (5.6) gives $R(b, u) \geq \sum_{j=0}^b q_j \rho^j$. For $0 < \theta \leq 1$ we have $\rho^j \geq (\theta \rho)^j$, hence

$$R(b, u) \geq \sum_{j=0}^b q_j (\theta \rho)^j = Q_u(\theta \rho) \mathbb{P}(J_\theta \leq b), \quad \mathbb{P}(J_\theta = j) = \frac{q_j (\theta \rho)^j}{Q_u(\theta \rho)}.$$

As in (5.3), $\mathbb{E}[z^{J_\theta}] = \prod_{m=0}^{u-1} ((1 - p_m)/(1 - p_m z))^{1/2}$ with $p_m = \theta m/u < 1$, so J_θ is a sum of independent negative-binomial variables of shape $\frac{1}{2}$, and $J_1 = J$. Such a variable with parameter p has mean $\frac{1}{2}p/(1-p)$ and variance $\frac{1}{2}p/(1-p)^2$, both increasing in p .

The bound on κ_K : $H_n \geq \log n + \gamma > \log n + \frac{1}{2}$, so $H_n > 2K + 3$ for every $n \geq e^{2K+5/2}$, and $\lceil e^{2K+5/2} \rceil < e^{2K+5/2} + 1 < e^{2K+3}$.

Write $\kappa = \kappa_K$. If $u < 4\kappa$, (18.5) follows from $R(b, u) \geq 1$, since then $\frac{u}{2} < 2\kappa$. Let $u \geq 4\kappa$ and $\theta = 1 - \kappa/u \in [\frac{3}{4}, 1)$, and index the factors by $k = u - m \in \{1, \dots, u\}$, so that $1 - m/u = k/u$ and $1 - \theta m/u = (k + \kappa(u - k))/u$.

- (i) By (5.1), $\log Q_u(\rho) - \log Q_u(\theta \rho) = \frac{1}{2} \sum_{k=1}^u \log(1 + \frac{\kappa(u-k)}{uk}) \leq \frac{\kappa}{2} \sum_{k=1}^u \frac{1}{k} = \frac{\kappa}{2} H_u$.
- (ii) $\text{Var } J_\theta \leq \text{Var } J \leq u^2$, by the monotonicity in p and (5.5).
- (iii) Termwise, $\frac{m}{u-m} - \frac{\theta m}{u-\theta m} = \frac{(1-\theta)m u}{(u-m)(u-\theta m)}$, so, using $\kappa(u-k)/u \leq \kappa$ in the denominator,

$$\mu - \mathbb{E}J_\theta = \frac{1}{2} \sum_{k=1}^u \frac{\kappa(u-k)}{k(k + \kappa(u-k)/u)} \geq \frac{u}{2} \sum_{k=1}^u \left(\frac{1}{k} - \frac{1}{k + \kappa} \right) - \frac{1}{2} \sum_{k=1}^u \frac{\kappa}{k + \kappa}.$$

The first sum telescopes to $H_\kappa - \sum_{k=u+1}^{u+\kappa} \frac{1}{k} \geq H_\kappa - \frac{\kappa}{u+1}$, and the second is at most $\kappa \int_0^u \frac{dx}{x+\kappa} = \kappa \log(1 + \frac{u}{\kappa})$. Since $x \log(1 + \frac{1}{x})$ increases with x and $x = \kappa/u \leq \frac{1}{4}$, $\frac{\kappa}{u+1} + \frac{\kappa}{u} \log(1 + \frac{u}{\kappa}) \leq \frac{1}{4} + \frac{1}{4} \log 5 < 1$. Hence $\mu - \mathbb{E}J_\theta \geq \frac{u}{2}(H_\kappa - 1)$.

Now $b \geq \mu - Ku$, (iii) and $H_\kappa \geq 2K + 3$ give $b - \mathbb{E}J_\theta \geq \frac{u}{2}(H_\kappa - 1) - Ku \geq u$. On $\{J_\theta > b\}$ we have $J_\theta - \mathbb{E}J_\theta > u$, so Cantelli's inequality (Section 5) and (ii) give

$\mathbb{P}(J_\theta > b) \leq u^2/(u^2 + u^2) = \frac{1}{2}$. With $\log Q_u(\rho) \geq \frac{u}{2} - \frac{1}{2} - \frac{1}{4} \log u$ (proof of Remark 18.1(2)), (i) and $H_u \leq 1 + \log u$,

$$\log R(b, u) \geq \frac{u}{2} - \frac{1}{2} - \frac{1}{4} \log u - \frac{\kappa}{2}(1 + \log u) - \log 2 \geq \frac{u}{2} - 2\kappa(1 + \log u).$$

(1) Put $K = \max\{0, \frac{1}{2}(H_u - 1) - \alpha\}$, so that (18.5) applies. By $H_u \leq 1 + \log u$, $2K + 3 \leq \max\{3, \log u - 2\alpha + 3\}$, so $\kappa_K \leq e^3 \max\{1, ue^{-2\alpha}\}$ and

$$2\kappa_K(1 + \log u) \leq 2e^3(1 + \log u)(1 + ue^{-2\alpha}) = o(u),$$

because $e^{-2\alpha} \log u = \exp\{-2(\alpha - \frac{1}{2} \log \log u)\} \rightarrow 0$. Hence $\log R(b, u) \geq \frac{u}{2} - o(u)$. Since $\alpha \rightarrow \infty$, the upper half follows from (18.2) exactly as in the proof of Remark 18.1(3). Both examples satisfy the hypothesis: $\alpha \geq \delta \log u$ directly, and $\alpha \geq \frac{1}{2}(H_u - 1) - K \geq \frac{1}{2} \log(u + 1) - \frac{1}{2} - K$ because $H_u \geq \log(u + 1)$.

(2) Put $L = \log N$ and $u = u_*(1 + \varepsilon)$ with $|\varepsilon| < 1/L$. By (5.8), $N/u_* = t$, $t = \log(4u_*) + 1 - \frac{r}{2}$ and $t + \log t = L + 1 + \log 4 - \frac{r}{2}$; so $t \sim L$, $t \leq L + 1 + \log 4$ once $t \geq 1$, $u \sim u_* \sim N/L$ and $\log N = O(\log u)$. Since $1/(1 + \varepsilon) \geq 1 - 1/L$,

$$\frac{N}{u} \geq t - \frac{t}{L} \geq \log(4u_*) - \frac{r}{2} - \frac{1 + \log 4}{L}, \quad \text{so} \quad \alpha = \frac{N/u - 1}{r} - \frac{1}{u} \geq \frac{\log(4u_*)}{r} - \frac{1}{2} - \frac{1}{r} - o(1).$$

With $H_u \leq 1 + \log u$ and $\log u \leq \log u_* + 1/L$,

$$\alpha - \frac{1}{2}(H_u - 1) \geq \left(\frac{1}{r} - \frac{1}{2}\right) \log u_* + \frac{\log 4 - 1}{r} - \frac{1}{2} - o(1),$$

which tends to $\log 2 - 1 > -1$ for $r = 2$ and to $+\infty$ for $r = 1$. So for large N the hypothesis holds with $K = 1$ ($\kappa_1 = 83$), and (18.5) gives the left inequality of (18.4). For the right one, the same lower bound for α gives $e^{-\alpha/3} = O(u_*^{-1/(3r)}) = O(u^{-1/(3r)})$, and $N/u \leq t/(1 - 1/L) \leq 2t$ gives $\alpha \leq 2t/r = O(\log u)$, so $(1 + \alpha)^{1/3} = O((\log u)^{1/3})$ and $E(b, u) = O(u^{1-1/(3r)}(\log u)^{1/3})$; the last term of (18.2) is $O(\log u)$ because $b < N$. $\square$

[write] 5 October 2026. Credit and checks. The idea is the independent check's (recorded at the end of Section 20): an exponential tilt $\theta = 1 - \kappa/u$ of the single-spine bound followed by Cantelli's inequality, with the estimates (i), (ii) and a weaker form of (iii), $\mu - \mathbb{E}J_\theta \geq \frac{u}{4}(H_\kappa - 1)$ for large u , and the conclusion $\log R(b, u) \geq \frac{u}{2} - O_K(\log u)$ for fixed K , which it tested numerically for $u \leq 120$. The write then wrote out the proof above, sharpened (iii) to $\frac{u}{2}(H_\kappa - 1)$ for all $u \geq 4\kappa$, made the constant explicit and uniform in K , and added consequences (1) and (2). Checks at the write, evidence only. The identity and the bound in (iii) hold exactly, in rationals, for all $4 \leq u \leq 40$ and $1 \leq \kappa \leq u/4$ (190 cases); in floating point, for all $4 \leq u \leq 1200$ and $1 \leq \kappa \leq u/4$ (179 700 cases), the least slack is 1.108 in (iii), at $u = 4$, and 0.610 in (i). For $u \leq 9$, $b \leq 48$ and every $0 \leq \kappa < u$, exact integers confirm $A(b, u) \geq [x^b]L_u \geq C_b u^{b+1} \sum_{j \leq b} q_j(\theta\rho)^j$ (2205 tilted cases). In floating point, for 90 values of $u \leq 580$, the single-spine ratio $[x^b]L_u/(C_b u^{b+1})$, which is at most $R(b, u)$, satisfies the last display of the proof wherever $b - \mathbb{E}J_\theta \geq u$ (19 922 cells, least slack 2.51) and (18.5) for $K = 0, \frac{1}{2}, 1, 2$ (18 247 cells); in the range $4\kappa_K \leq u \leq 600$, where (18.5) does not follow from $R \geq 1$ ($\kappa_K = 11, 31, 83$ for $K = 0, \frac{1}{2}, 1$; 6195 cells), its least slack is 106, 362 and 1131. The bound is thus far from sharp at these sizes; it is a statement about the order $O_K(\log u)$, not a numerical estimate. In the window of (2) for $r = 2$, the least value of $\alpha - \frac{1}{2}(H_u - 1)$ over the admissible u , with exact H_u , lies between -0.080 and $+0.032$ for nine values of N from 40 to 10^6 (-0.058 at $N = 10^6$), far above the -1 that the proof uses.

Other substantive questions remain:

1. What are the centered fluctuation scales and limiting laws of U_n and H_n ? A large-deviation principle for fixed relative deviations does not by itself prove a central limit theorem.

2. How large is $U_n - H_n$ on its typical, sub- $n/\log n$ scale? The fixed-deficit rate is informative but does not answer this local question.
3. What fraction of a longest unary path is a consecutive initial root chain? The path concentration theorem does not identify that statistic.
4. Can the weighted-placement sum (12.3) be controlled uniformly to successive orders, leading to a genuine expansion and a correspondingly sharper inverse?

These are mathematically meaningful continuations; no answer is inferred from finite numerical agreement or from expanding the Lambert lower envelope alone.

[write] **5 October 2026.** Status after the merge. Question 1 is open (Section 20, item 1). Question 2 is Part I's question 2; both are merged into item 2 of Section 20. Question 3 is open (item 3 there). Question 4 is partly answered by Part I: every fixed inverse-logarithmic order of the logarithm of the total count is determined (Theorem 1.1); uniform control of (12.3) itself, and the multiplicative level, remain open (merged with Part I's question 3 in Section 9.2, item 3).

19 Reproducibility and finite validation

The source package contains this T_EX file, its PDF, an exact checker, integer data tables, integrity and corruption tests, and deterministic build instructions. It contains no downloaded source-paper PDFs. The bibliography links the primary sources instead.

The executable mathematical checker uses only the Python standard library, integers, and `fractions.Fraction`. Every test uses an explicit conditional exception; no validation depends on `assert`, so Python's `-O` mode does not disable checks. The suite performs the following logically independent finite comparisons:

1. Reconstruct both total-count sequences and first unary moments through $n = 200$ by the size recurrence. Check monotonicity, moment ranges, and the total skeleton sandwich.
2. Compare the first 16 terms of each model with the published prefixes. The complete remote OEIS b-files were not successfully retrieved, so the longer computed tables are not claimed to have been checked against them.
3. Compare totals and first moments through $n = 20$ with a bivariate decomposition indexed by unary and binary counts.
4. Compare that direct decomposition with the coefficient form of the radical recurrence. Check an exact integer upper corollary using $K < 12$, for $1 \leq u \leq 9$ and $0 \leq b \leq 30$.
5. Enumerate every binary skeleton and unary-slot composition for $0 \leq b \leq 4$, $1 \leq u \leq 6$, accumulating exact products of leaf depths by height. Verify the total and all applicable joint upper and constructive lower bounds.
6. Compute the maximum squared radical product dynamically through $u = 75$ and compare it with $u^{2u-1}/(u!)^2$ using rational arithmetic.
7. Compare two exact reduced-shape counts through $u = 100$: the quadratic recurrence and

$$d_u = \sum_{q=0}^{u-1} 2^{-q} C_q \binom{u+q-1}{2q}.$$

Also check the weaker rational-radius bound $d_u \leq 4^u/2$.

The last formula follows by contracting the unary chains in a reduced shape with q binary vertices. Its $q + 1$ leaf slots require at least one unary vertex each, while its other slots may be empty. A weak composition of the remaining $u - q - 1$ vertices into $2q + 1$ slots gives $\binom{u+q-1}{2q}$ possibilities.

These tests check finite instances and implementation consistency. They do not prove the asymptotic theorems, certify an error constant, establish an eventual numerical threshold, or prove global novelty. The proofs above are independent of the data.

[write] 5 October 2026. Item 7’s formula, and the paragraph after the list, are Part I’s Lemma 2.1 with the same proof; Report 110 states it as a lemma.

Run the full suite from the extracted package with

```
python3 check.py
python3 -O check.py
python3 integrity.py
python3 corruption_test.py
python3 build.py
```

The build requires a standard `pdflatex` installation and the listed $\text{\TeX}$ packages. It fixes the PDF time and identifier inputs, runs multiple passes, rejects unresolved references and overfull boxes, and checks reproducibility by a clean second build. The package manifest records SHA-256 hashes of the distributable files. Re-extraction and rebuild provide a separate packaging check. Page images are reviewed for every page of the final PDF; integrity and reproducibility are distinct from mathematical validity.

[write] 5 October 2026. In this repository the files named above are shipped under other names (`code/109-binding-check.py`, `code/109-binding-validation-auditor_probes.py`, `data/109-binding-A135501_computed.txt` and so on; the report’s `README.md` maps them), and the scripts expect the delivered layout, so they run only on a copy of the delivered package. The delivered `report109.tex`, `report109.pdf`, `manifest.json` and `README.md` are not shipped (the text is printed here as Part II). That `README`’s sentence that “the coefficient of $n/\log n \dots$ remain[s] unresolved here” is true of Report 109 and stale for this report: Part I determines the coefficient; the amplitude, equivalent, transseries and fluctuation laws it lists remain open.

20 Further questions and research

[write] (5 October 2026.) Under Vladimir’s standing rule of 4 October 2026, this section collects every claim of this Part that is stated without proof, with its source, the argument sketched for it and what is missing, and records how Part I and Remark 18.1 re-scope it. Nothing in this Part was found to be wrong; its stale statements carry dated notes in place. The questions on the count itself (equivalent, amplitude, residual scale, the weighted-placement sum, constants, literature) are in Section 9.2.

1. *Fluctuations of U_n and H_n* (Section 18, question 1): the centred fluctuation scales and limit laws. Theorem 11.3 concerns fixed relative deviations at speed n and does not give a central limit theorem. Missing: a local analysis below the scale $n/\log n$. The rate of the L^p convergence is not stated either; evidence only, from the shipped first moments (`data/109-binding-A135501_unary_moment.txt` and the A220894 file), computed at the write: $\mathbb{E}U_n \log N/N$ is 1.217, 1.245, 1.252 for $s = 1$ and 1.015, 1.018, 1.018 for $s = 0$ at $n = 50, 100, 200$, still well above 1 for $s = 1$.
2. *The typical size of $U_n - H_n$* (merged: this Part’s question 2 and Part I’s question 2). Report 109 asks how large $U_n - H_n$ is on its typical, sub- $n/\log n$ scale; Report 110 asks for the typical off-spine deficit $u - h$ near the moving saddle, where h , the largest

unary depth of a terminal leaf of the reduced tree, is H_n for an actual term. Known: Corollary 11.4 gives the deficit rate d/r at speed n and $U_n - H_n = o(n/\log n)$ in probability. Report 110 adds that the deficit d optimizing its majorant (Proposition 4.2) is not a distributional law. Missing: any estimate of the typical order.

3. *The fraction of a longest unary path that is a consecutive initial root chain* (question 3). Corollary 11.4 is about a root-to-leaf path and does not identify this statistic.
4. *Uniform control of the weighted-placement sum* (question 4): merged with Part I's question 3, Section 9.2, item 3, where its partial answer is recorded.
5. *The ratio frontier beyond Remark 18.1* (added by the write). Open: the second-order term of $\log R(b, u) - \frac{u}{2}$ at the moving saddle (known to lie between $-O(\log u)$ and $O(u^{1-1/(3r)}(\log u)^{1/3})$; for single (b, u) this is the analogue of Section 9.2, item 2), and $\lim \log R(b, u)/u$ for bounded $\alpha = b/u$, or for $\alpha \rightarrow \infty$ with $\alpha - \frac{1}{2}(H_u - 1)$ not bounded below by a positive constant. Known there: $0 \leq \log R(b, u) \leq \frac{u}{2} + E(b, u) + O(\log(u + b + 1))$, so $\limsup \log R/u \leq \frac{1}{2} + 3 \cdot 2^{1/3}(1 + \alpha)^{1/3}e^{-\alpha/3}$ at fixed α . Sketch: the single-spine lower bound needs $\mathbb{P}(J \leq b)$ in a lower-tail regime of the negative-binomial sum J , which Cantelli's inequality does not reach. (Dated note, 5 October 2026: re-scoped after the independent check. Proposition 18.2 reaches most of that regime: tilting J by $\theta = 1 - \kappa/u$ before applying Cantelli's inequality gives $\log R(b, u) \geq \frac{u}{2} - 2\kappa_K(1 + \log u)$ whenever $\alpha \geq \frac{1}{2}(H_u - 1) - K$, with $\kappa_K \leq e^{2K+3}$, and hence the limit $\frac{1}{2}$ whenever $\alpha - \frac{1}{2} \log \log u \rightarrow \infty$ and $\log(b + 1) = o(u)$, in particular in all of Part I's window for $r = 2$. The sketch's "which Cantelli's inequality does not reach" is true of J itself, not of the tilted variables. Open now: the second-order term, as above, and $\lim \log R(b, u)/u$ when $\alpha - \frac{1}{2} \log \log u$ does not tend to infinity, in particular for bounded α . There neither bound gives $\frac{1}{2}$: for bounded α , $E(b, u)/u$ does not tend to 0, and the tilt that the proof needs, κ of order $ue^{-2\alpha}$, costs $\frac{\kappa}{2}H_u$, which is not $o(u)$.)

This Part's other limitations are printed in place and are not questions: the comparison with David et al. does not subsume their normalization or head-abstraction theorems; natural-size, linear, affine, beta-normal, typed, bounded-index and bounded-height classes are not substitutes; only 16 OEIS prefix terms per model were compared, the remote b-files were not retrieved, and the longer tables are internally computed; **-fixtures-only** is a prefix check only; finite checks prove no asymptotic statement.

Independent check of the write (dated note, 5 October 2026). An independent adversarial check of the write, made by the intake after the write (b432720bf), reread the eleven items of the write that carry mathematics: the four parts of Remark 18.1 with their proofs; the note after it (its scope, the part of the window not covered for $r = 2$, its evidence); the coefficient $\frac{1}{2} - \frac{\log 4}{r}$ of $n/\log n$ and $M_r(N) - \frac{N}{r}(w - 2 + \frac{1}{w}) = \frac{N}{2 \log N}(1 + o(1))$ (note after Theorem 11.1); the attained residual factor (note after the remark "What exponential precision is available" of Section 14); the constant $1 + \frac{r}{2} - 2 \log(4r)$ that Part II's inverse omits, the exact order of its error and the side on which it lies (note after Theorem 1.2); the gap between the two saddles u_* (front matter, notation); the front-matter row on the ratio; and item 5 above. It found every mathematical claim valid, no counterexample and no gap in any proof chain. Three sentences said more than, or less precisely than, their mathematics; each is corrected where it stands, with a dated note keeping the first wording. The note after the remark on exponential precision did not say against which form of Theorem 11.1 the residual is measured: the constant is $\frac{1}{2}$ against the Lambert form and $\frac{1}{2} - \frac{\log 4}{r}$ against the four-scale form that the remark displays. The evidence sentence after Remark 18.1 called the full ratios decreasing, which 0.737, 0.738, 0.732 are not. The front-matter row claimed

two-sided bounds for all (b, u) , while the lower bound $\frac{u}{2} - O(\log u)$ needs $b \geq \frac{u}{2}(H_u - 1) + au$. The check also proposed an exponential tilt of the single-spine bound; the write proved it in full as Proposition 18.2, which closes the uncovered part of the window, and re-scoped item 5. No other statement or number changed.

The check used neither the delivered programs nor the write's. Its own recurrence for $A_s(n)$, from $L_m = mz^s + zL_{m+1} + zL_m^2$ in exact integers, matched the OEIS b-files (fetched 5 October 2026) for A135501 at every $n \leq 300$ and for A220894 at every $n \leq 200$, with no mismatch, and was run to $n = 600$; summing $A(b, u)$ over the admissible u recovered $A_s(n)$ exactly for $n \leq 24$ in both models. It recomputed $A(b, u)$ from (2.3), exactly for $u \leq 8$, $b \leq 400$ and for $u \leq 24$, $b \leq 44$, and in scaled floating point for $u \leq 120$, $b \leq 400$, agreeing with the exact values to $6 \cdot 10^{-14}$ in the logarithm. On all these cells (4.7) and (18.2) hold, with least slack 5.67 in the logarithm; (18.3), with the best admissible a at each cell, holds on all 29 160 floating-point cells and on all exact cells with $u \leq 8$, tightest ($6 \cdot 10^{-6}$) at $u = 1$, $b = 400$, where $R = 1$ and the bound tends to 0. It re-derived every step of the proof of Remark 18.1, and of Proposition 4.2 itself, and measured the Cantelli margin $\alpha - \frac{1}{2}(H_u - 1)$, with exact H_u , at the admissible u nearest u_* : for $r = 2$ it lies between 0.197 (at $N = 72$) and 0.43 for $20 \leq N \leq 740$, above $a_0 = 0.0966$ throughout, and for $r = 1$ it grows from 1.4 to 3.2. At $N = 740$ and $r = 2$ the margin is +0.391 at $u = 120$, +0.014 at $u = 134$, -0.034 at $u = 136$ and -0.080 at $u = 138$, inside the window $|u/u_* - 1| < 1/\log N$ (edge 138.0): Remark 18.1(4) does not reach those u , as the note after it says. It reproduced the write's evidence values 0.737, 0.738, 0.732 and 0.576, 0.568, 0.559 exactly. At 160 digits, with $N = e^L$ and $\ell = \log L$, the residual $[rM_r(N)/N - (L - \ell + \log 4 - 1)]L - (\ell - \log 4 + \frac{r}{2})$ of the coefficient expansion is $3.7 \cdot 10^{-2}$, $2.7 \cdot 10^{-3}$, $1.4 \cdot 10^{-6}$ and $6 \cdot 10^{-14}$ at $L = 10^2$, 10^4 , 10^8 and 10^{16} , and $(M_r(N) - \frac{N}{r}(w - 2 + \frac{1}{w}))/((N/(2L)) - 1)$ falls to $3.5 \cdot 10^{-15}$ at $L = 10^{16}$. Solving $M_r(N_0) = Y$ through the parametric form, the omitted inverse constant is recovered as -1.272589 ($r = 1$) and -2.158883 ($r = 2$) at $v = 10^{15}$, and the ratio of Part II's shortfall to $r|1 + \frac{r}{2} - 2 \log(4r)|Y/v^3$ is 1.000000 there ($r = 1$). It also confirmed $w - t \rightarrow \frac{r}{2}$ (0.999999 at $L = 10^6$ for $r = 2$). This was a careful reading with numerical tests, not a formal verification.

References

- [1] OEIS Foundation (Report 109: OEIS Foundation Inc.), *A135501, Number of closed lambda-terms of size n* . <https://oeis.org/A135501>. Report 109: model and displayed prefix checked 2 October 2026.
- [2] OEIS Foundation (Report 109: OEIS Foundation Inc.), *A220894, Number of closed lambda-terms of size n when variables have size zero* (Report 109's title: *... of size n with variables of size zero*). <https://oeis.org/A220894>. Report 109: model and displayed prefix checked 2 October 2026.
- [3] O. Bodini, D. Gardy, B. Gittenberger, and A. Jacquot, *Enumeration of generalized BCI lambda-terms*, Electronic Journal of Combinatorics **20**(4), P30 (2013). Section 5, Theorem 22 and Remark 23 (Report 109). <https://www.combinatorics.org/ojs/index.php/eljc/article/view/v20i4p30>; PDF: <https://www.combinatorics.org/ojs/index.php/eljc/article/download/v20i4p30/pdf/>.
- [4] R. David, K. Grygiel, J. Kozik, C. Raffalli, G. Theyssier, and M. Zaionc, *Asymptotically almost all lambda-terms are strongly normalizing*, Logical Methods in Computer Science **9**(1:2) (2013). [https://doi.org/10.2168/LMCS-9\(1:2\)2013](https://doi.org/10.2168/LMCS-9(1:2)2013).
- [5] O. Bodini, D. Gardy, B. Gittenberger, and Z. Gołębiewski, *On the number of unary-binary tree-like structures with restrictions on the unary height*, Annals of Combinatorics **22**(1), 45–91 (2018). Expanded preprint (2015): <https://arxiv.org/abs/1510.01167>.
- [6] B. Gittenberger and I. Larcher, *Distribution of variables in lambda-terms with restrictions on De Bruijn indices and De Bruijn levels*, Electronic Journal of Combinatorics **26**(4), P4.47 (2019). <https://arxiv.org/abs/1903.05243>.
- [7] K. Grygiel and I. Larcher, *Unary profile of lambda terms with restricted De Bruijn indices*, Discrete Mathematics & Theoretical Computer Science **22**:3, article 2 (2021). <https://dmtcs.episciences.org/7135/pdf>.
- [8] M. Bendkowski, O. Bodini, and S. Dovgal, *Statistical properties of lambda terms*, Electronic Journal of Combinatorics **26**(4), P4.1 (2019). <https://www.combinatorics.org/ojs/index.php/eljc/article/download/v26i4p1/pdf/>.

[**write**] Bibliography merged from the two manuscripts (5 October 2026). Entries [1]–[5] are in both, under Report 110's keys; [6], [7] and [8] are Report 109's only. The citations were not checked against the publications at the placement or at this write.